

# Medium Access Control Sublayer

## Chapter 4

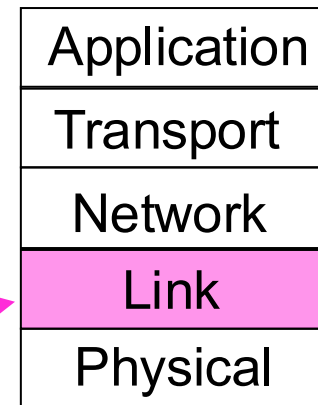
- Channel Allocation Problem
- Multiple Access Protocols
- Ethernet
- Wireless LANs
- Broadband Wireless
- Bluetooth
- RFID
- Data Link Layer Switching

Revised: August 2011

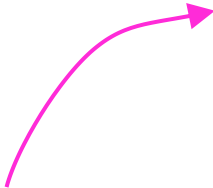
# The MAC Sublayer

Responsible for deciding who sends next on a multi-access link

- An important part of the link layer, especially for LANs



MAC is in here!



# Channel Allocation Problem (1)

For fixed channel and traffic from  $N$  users

- Divide up bandwidth using FTM, TDM, CDMA, etc.
- This is a static allocation, e.g., FM radio

This static allocation performs poorly for bursty traffic

- Allocation to a user will sometimes go unused

# Channel Allocation Problem (2)

Dynamic allocation gives the channel to a user when they need it. Potentially N times as efficient for N users.

Schemes vary with assumptions:

| Assumption                 | Implication                                  |
|----------------------------|----------------------------------------------|
| Independent traffic        | Often not a good model, but permits analysis |
| Single channel             | No external way to coordinate senders        |
| Observable collisions      | Needed for reliability; mechanisms vary      |
| Continuous or slotted time | Slotting may improve performance             |
| Carrier sense              | Can improve performance if available         |

# Multiple Access Protocols

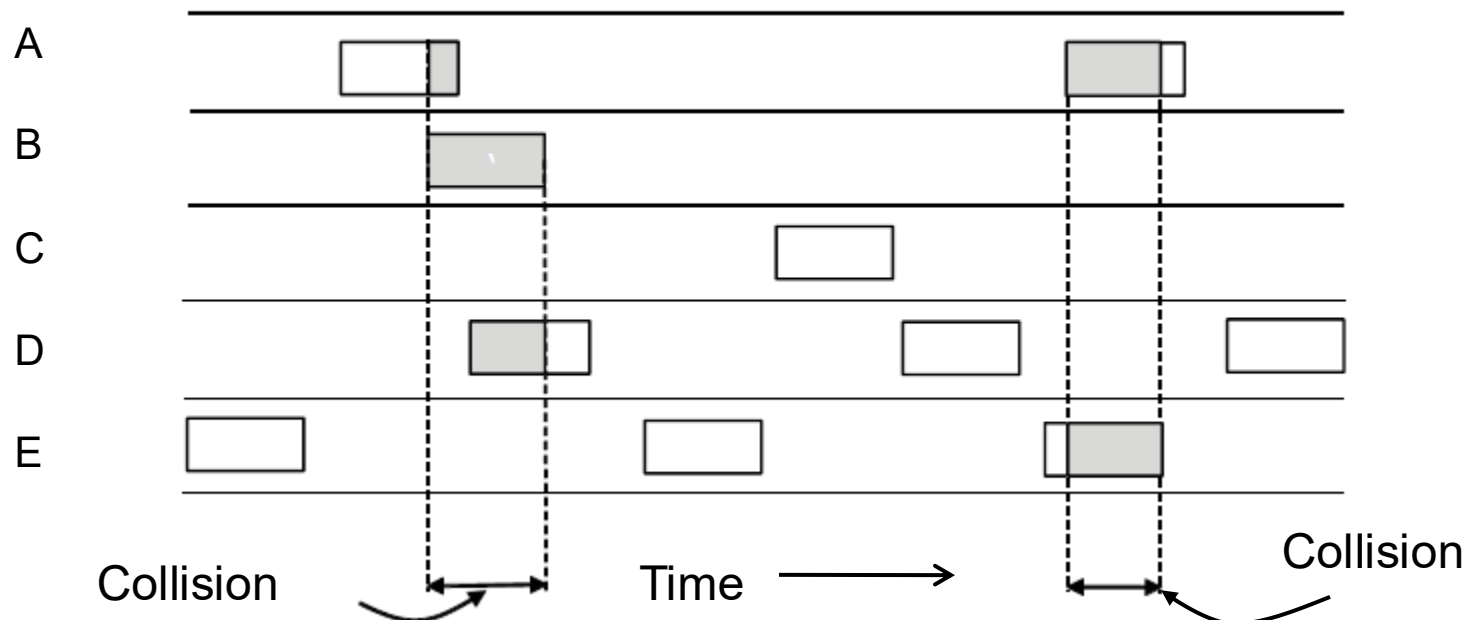
- ALOHA »
- CSMA (Carrier Sense Multiple Access) »
- Collision-free protocols »
- Limited-contention protocols »
- Wireless LAN protocols »

# ALOHA (1)

In pure ALOHA, users transmit frames whenever they have data; users retry after a random time for collisions

- Efficient and low-delay under low load

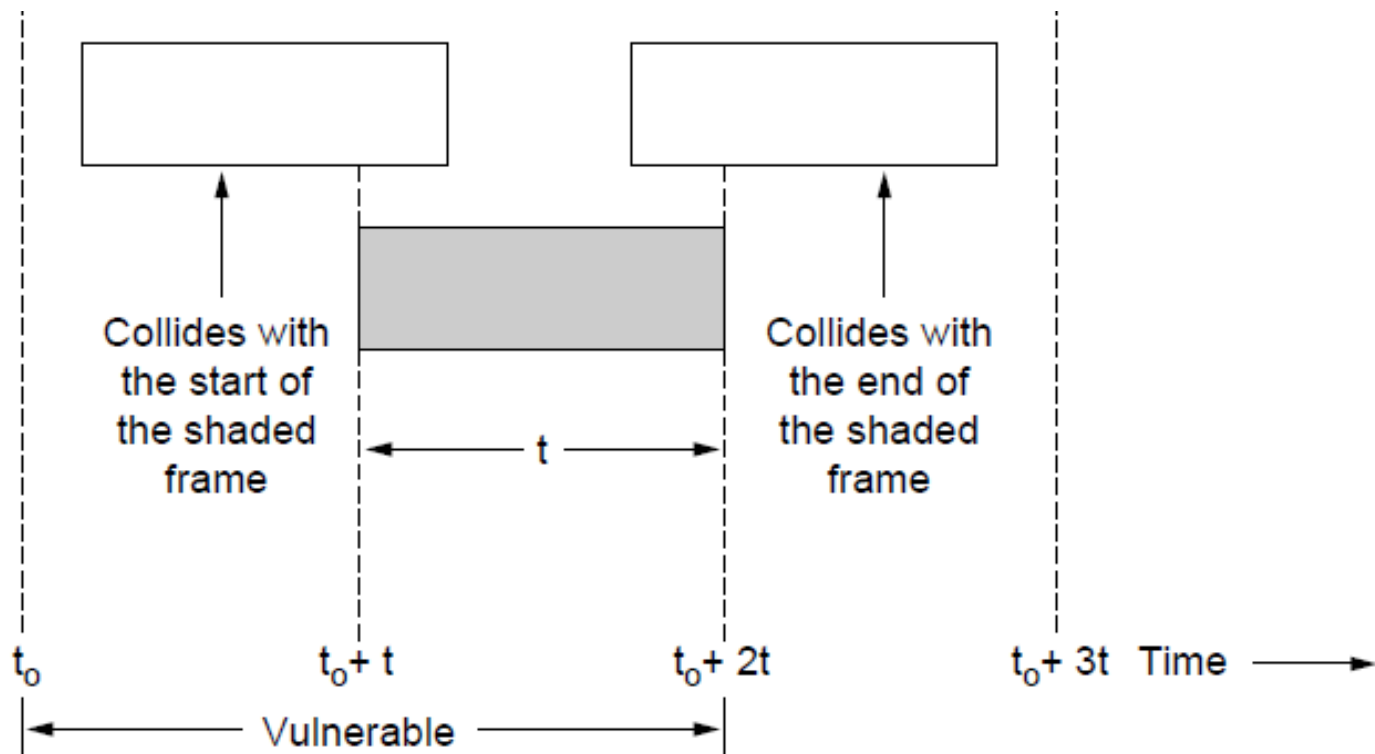
User



# ALOHA (2)

Collisions happen when other users transmit during a vulnerable period that is twice the frame time

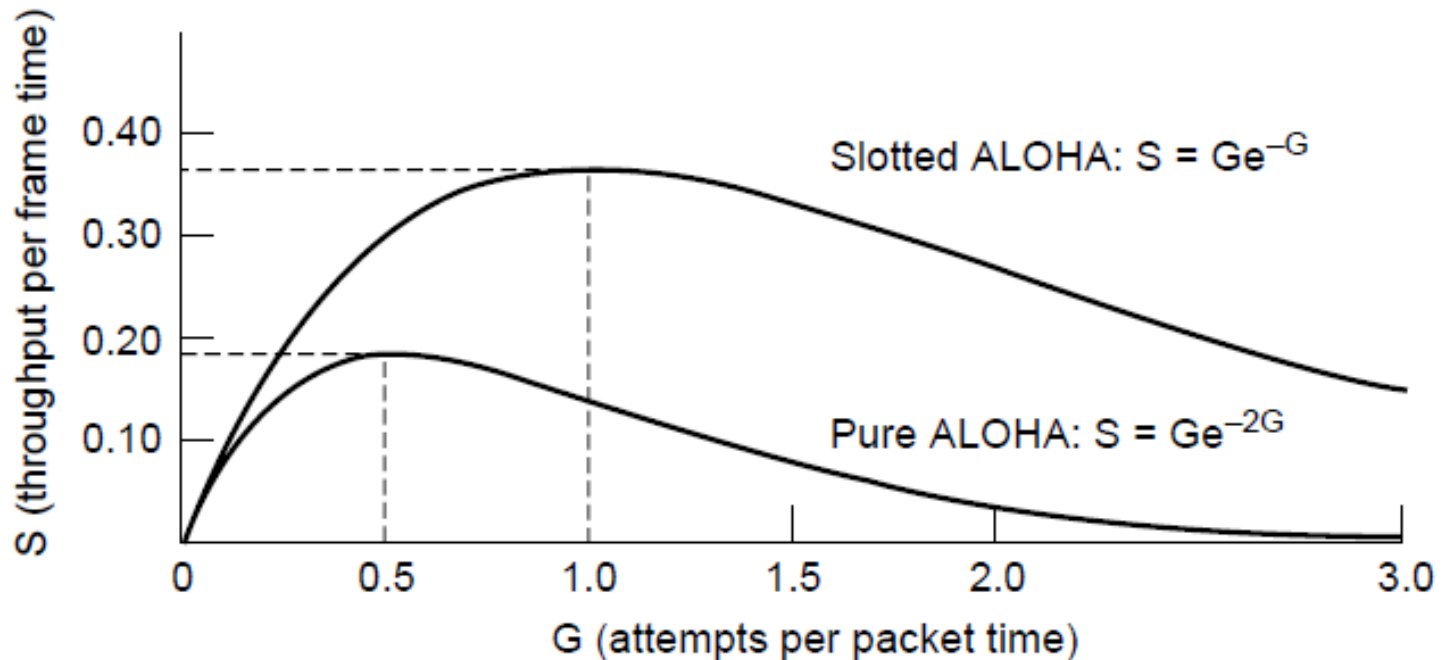
- Synchronizing senders to slots can reduce collisions



# ALOHA (3)

Slotted ALOHA is twice as efficient as pure ALOHA

- Low load wastes slots, high loads causes collisions
- Efficiency up to  $1/e$  (37%) for random traffic models





# CSMA (1)

CSMA improves on ALOHA by sensing the channel!

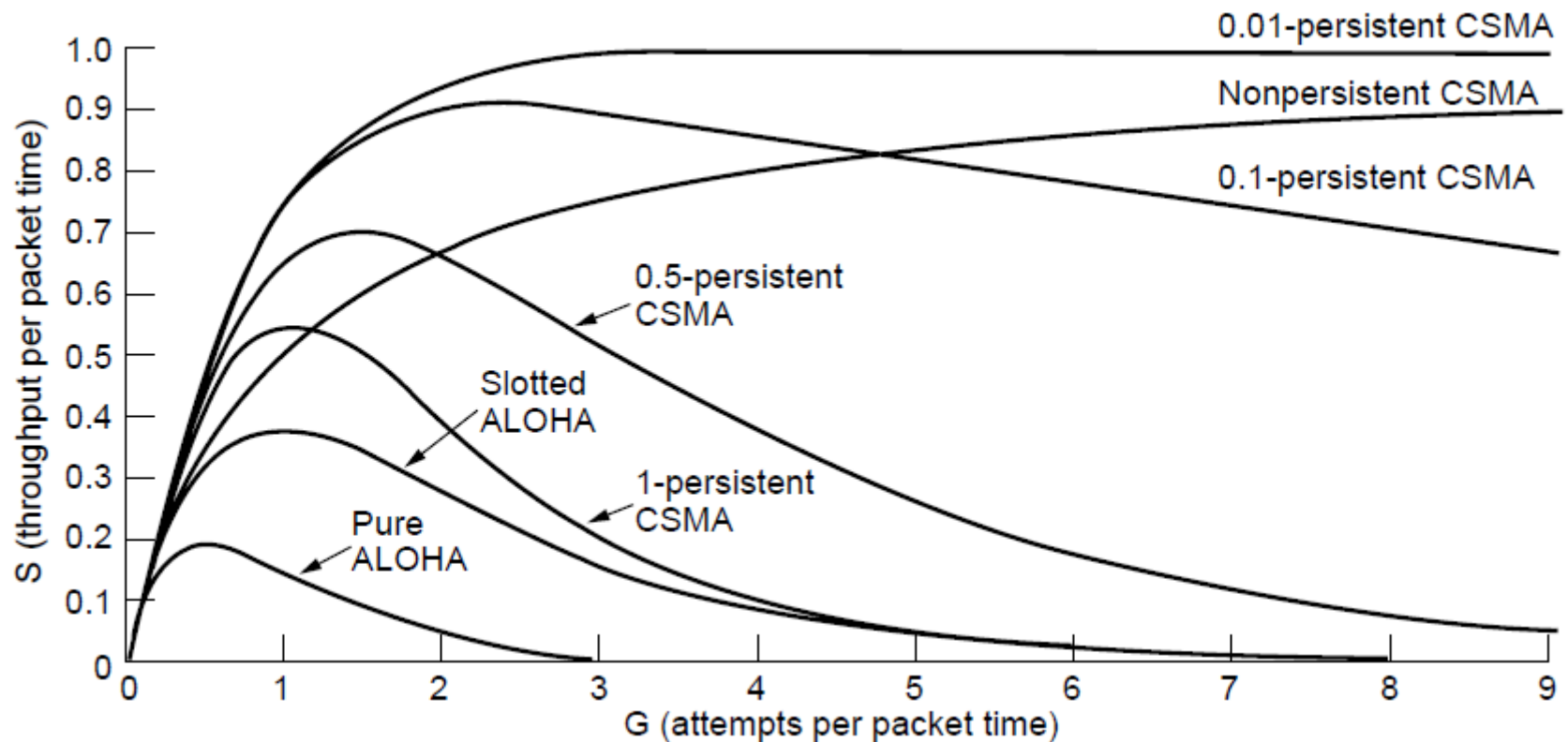
- User doesn't send if it senses someone else

Variations on what to do if the channel is busy:

- 1-persistent (greedy) sends as soon as idle
- Nonpersistent waits a random time then tries again
- p-persistent sends with probability  $p$  when idle

# CSMA (2) – Persistence

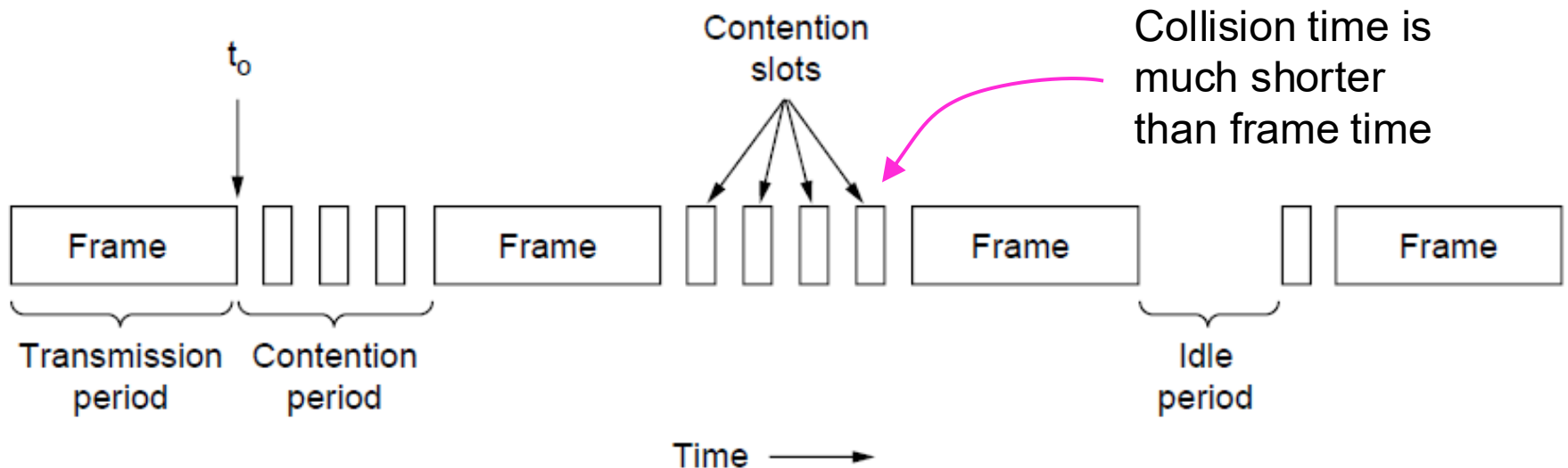
CSMA outperforms ALOHA, and being less persistent is better under high load



# CSMA (3) – Collision Detection

CSMA/CD improvement is to detect/abort collisions

- Reduced contention times improve performance



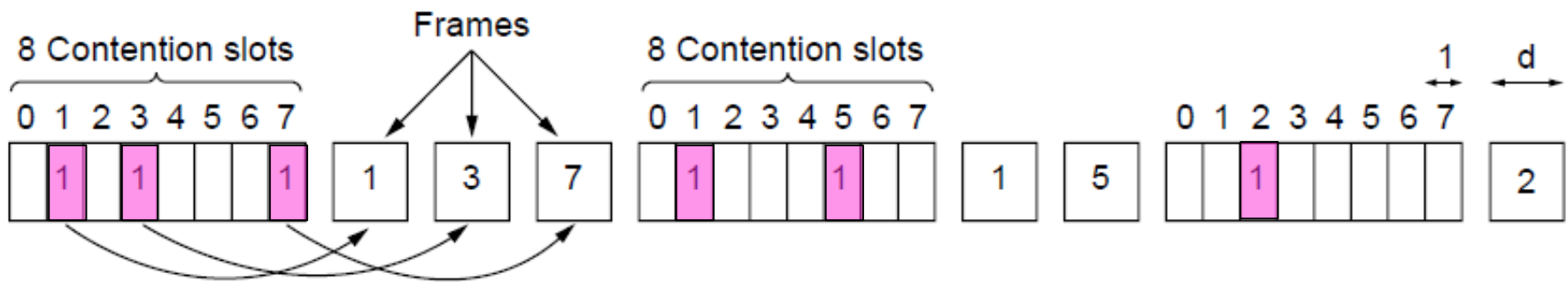
# Collision-Free (1) – Bitmap

Collision-free protocols avoid collisions entirely

- Senders must know when it is their turn to send

The basic bit-map protocol:

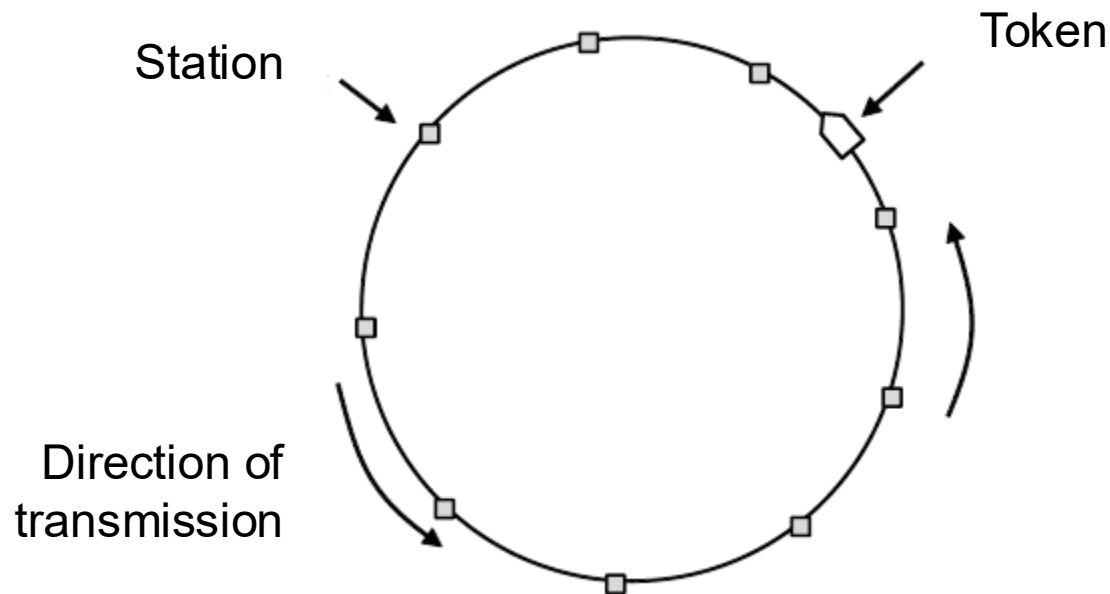
- Sender set a bit in contention slot if they have data
- Senders send in turn; everyone knows who has data



# Collision-Free (2) – Token Ring

Token sent round ring defines the sending order

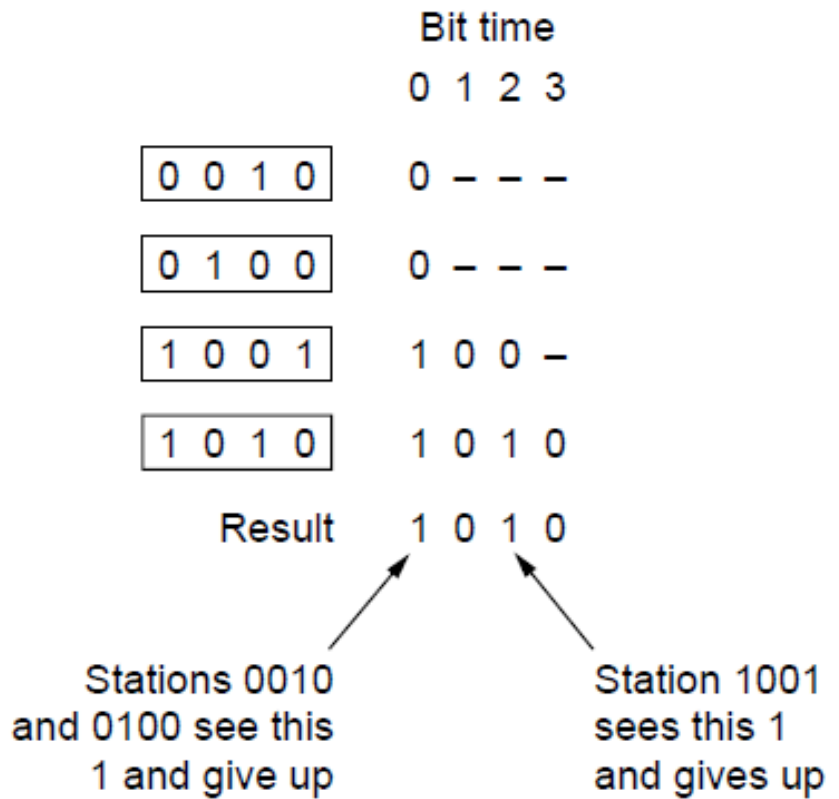
- Station with token may send a frame before passing
- Idea can be used without ring too, e.g., token bus



# Collision-Free (3) – Countdown

Binary countdown improves on the bitmap protocol

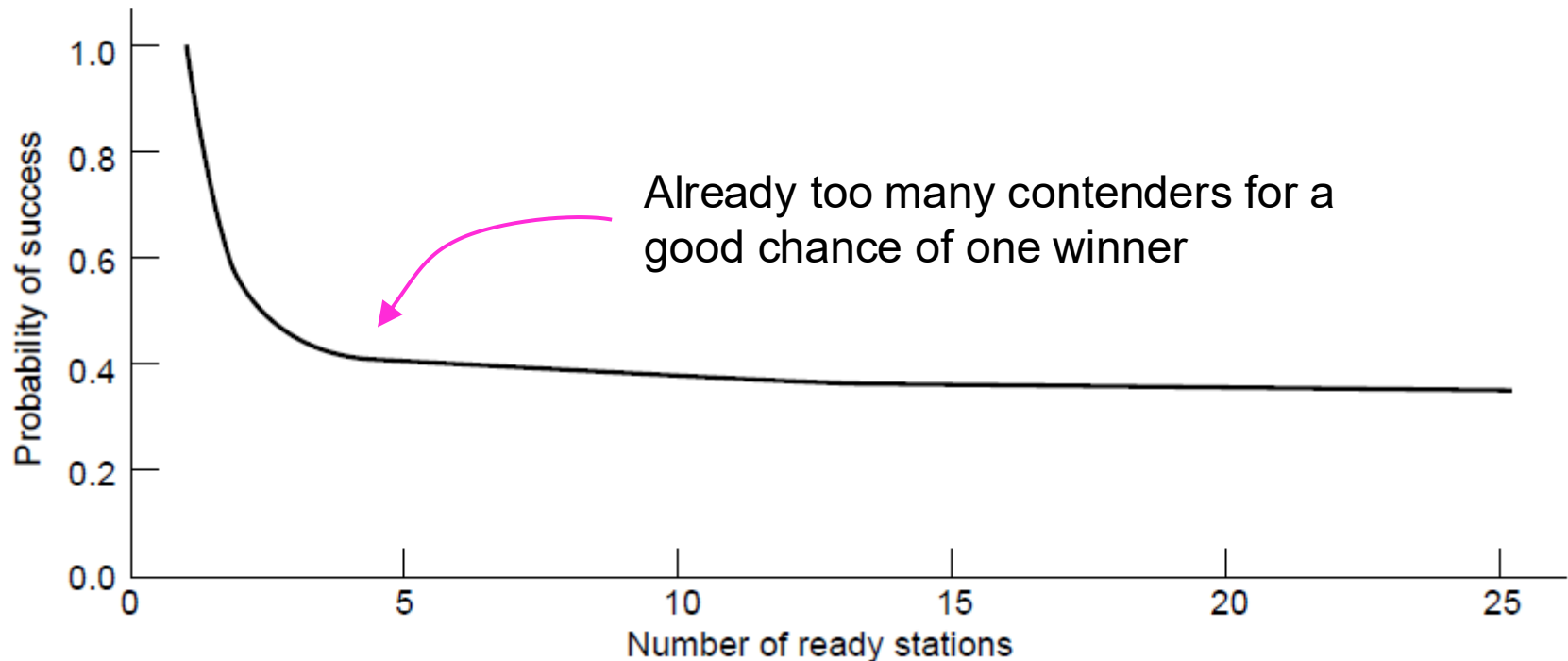
- Stations send their address in contention slot (log N bits instead of N bits)
- Medium ORs bits; stations give up when they send a “0” but see a “1”
- Station that sees its full address is next to send



# Limited-Contention Protocols (1)

Idea is to divide stations into groups within which only a very small number are likely to want to send

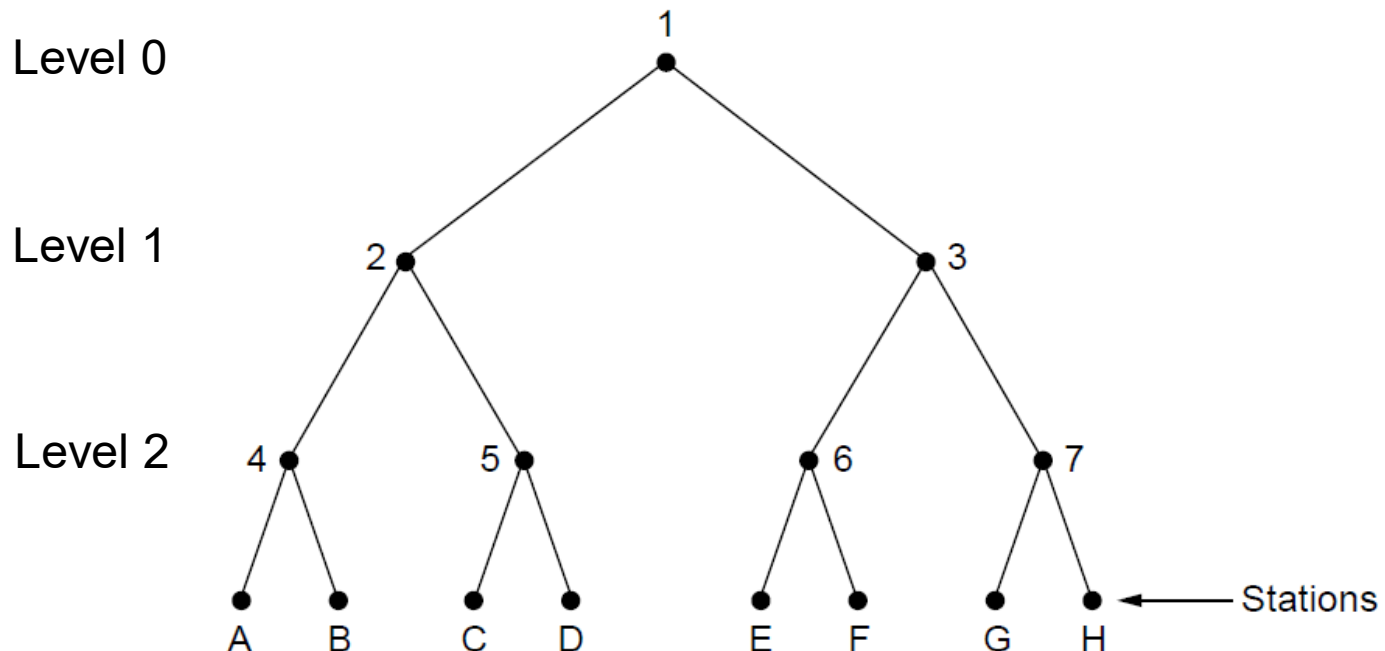
- Avoids wastage due to idle periods and collisions



# Limited Contention (2) – Adaptive Tree Walk

Tree divides stations into groups (nodes) to poll

- Depth first search under nodes with poll collisions
- Start search at lower levels if  $>1$  station expected





# Wireless LAN Protocols (1)

Wireless has complications compared to wired.

Nodes may have different coverage regions

- Leads to hidden and exposed terminals

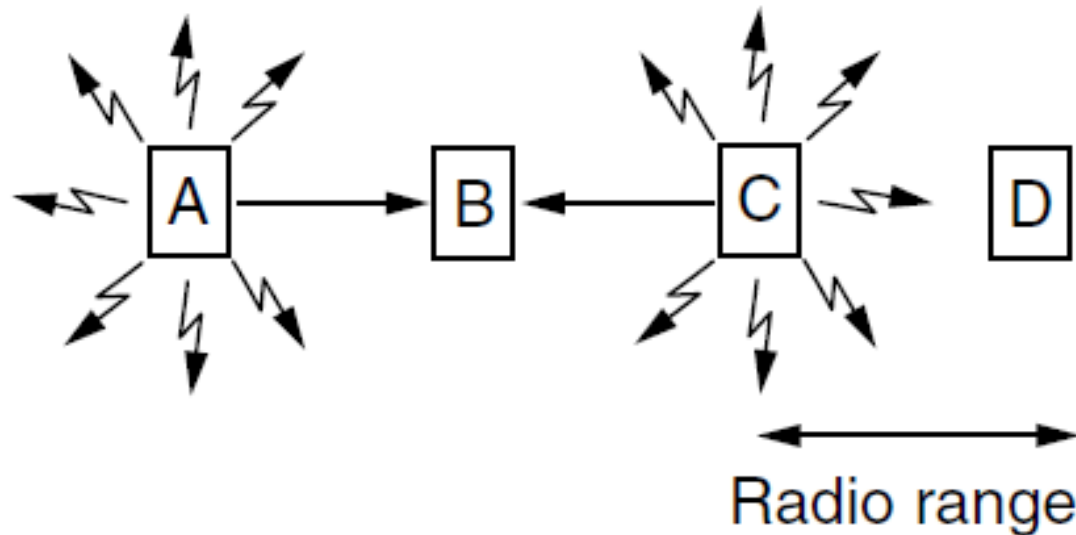
Nodes can't detect collisions, i.e., sense while sending

- Makes collisions expensive and to be avoided

# Wireless LANs (2) – Hidden terminals

Hidden terminals are senders that cannot sense each other but nonetheless collide at intended receiver

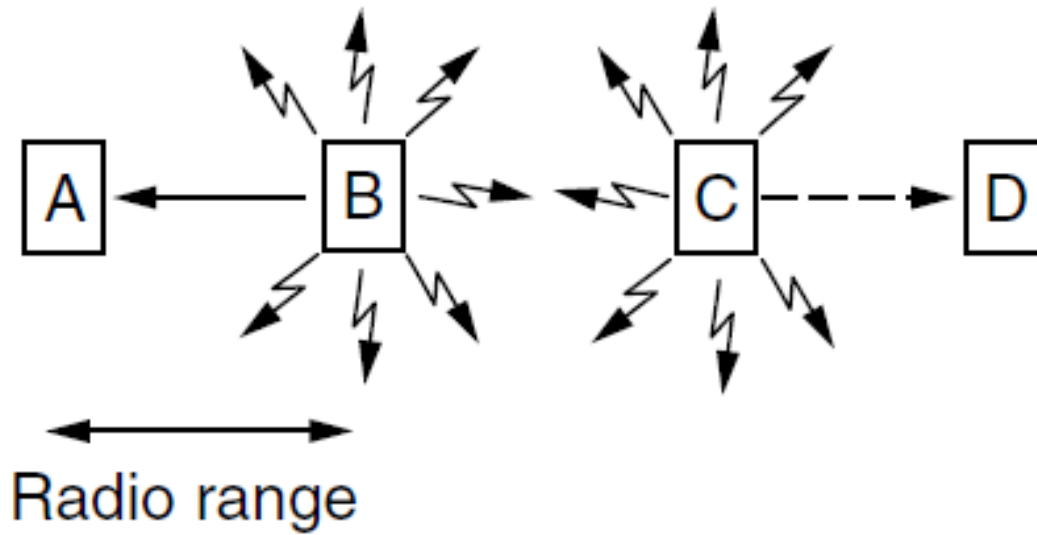
- Want to prevent; loss of efficiency
- A and C are hidden terminals when sending to B



# Wireless LANs (3) – Exposed terminals

Exposed terminals are senders who can sense each other but still transmit safely (to different receivers)

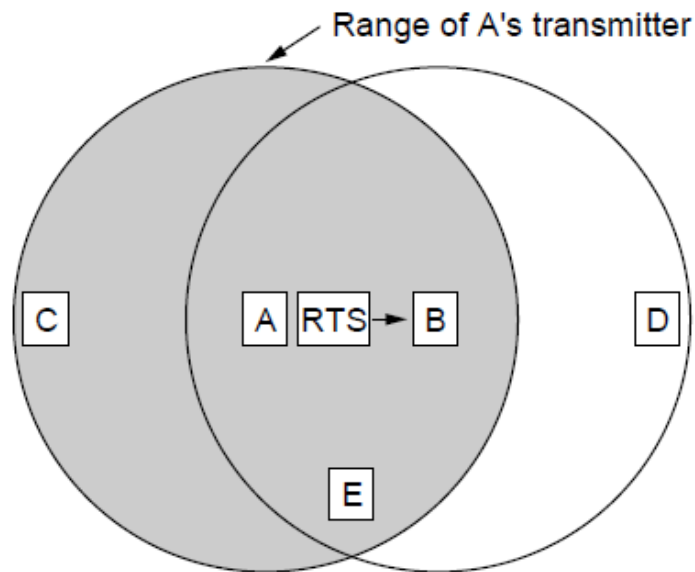
- Desirably concurrency; improves performance
- $B \rightarrow A$  and  $C \rightarrow D$  are exposed terminals



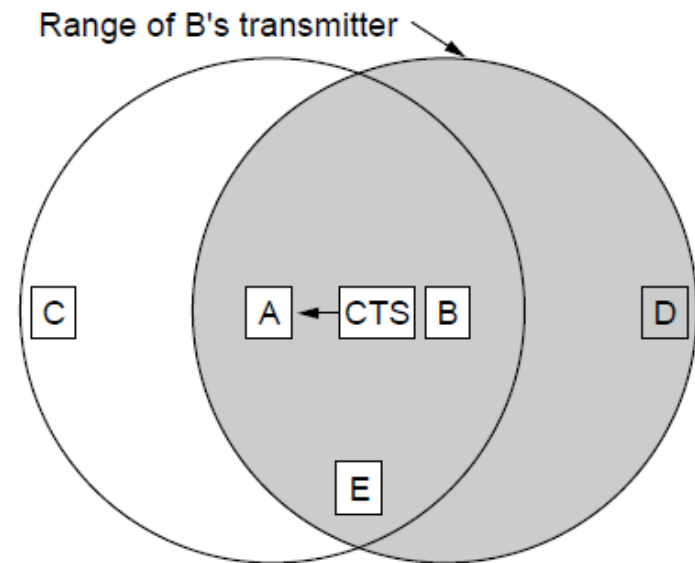
# Wireless LANs (4) – MACA

MACA protocol grants access for A to send to B:

- A sends RTS to B [left]; B replies with CTS [right]
- A can send with exposed but no hidden terminals



A sends RTS to B; C and E  
hear and defer for CTS



B replies with CTS; D and  
E hear and defer for data

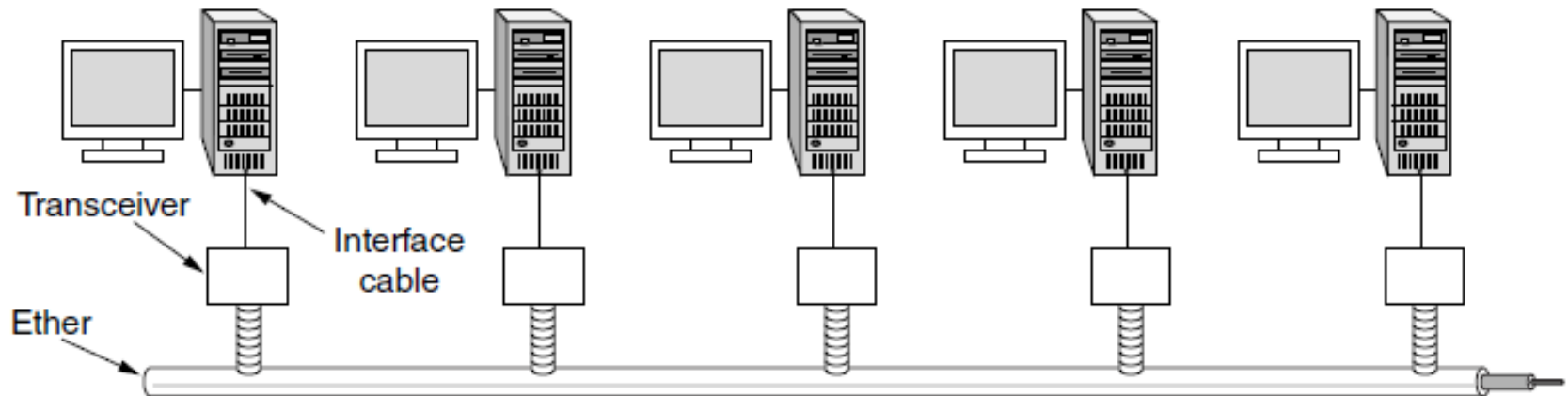
# Ethernet

- Classic Ethernet »
- Switched/Fast Ethernet »
- Gigabit/10 Gigabit Ethernet »

# Classic Ethernet (1) – Physical Layer

One shared coaxial cable to which all hosts attached

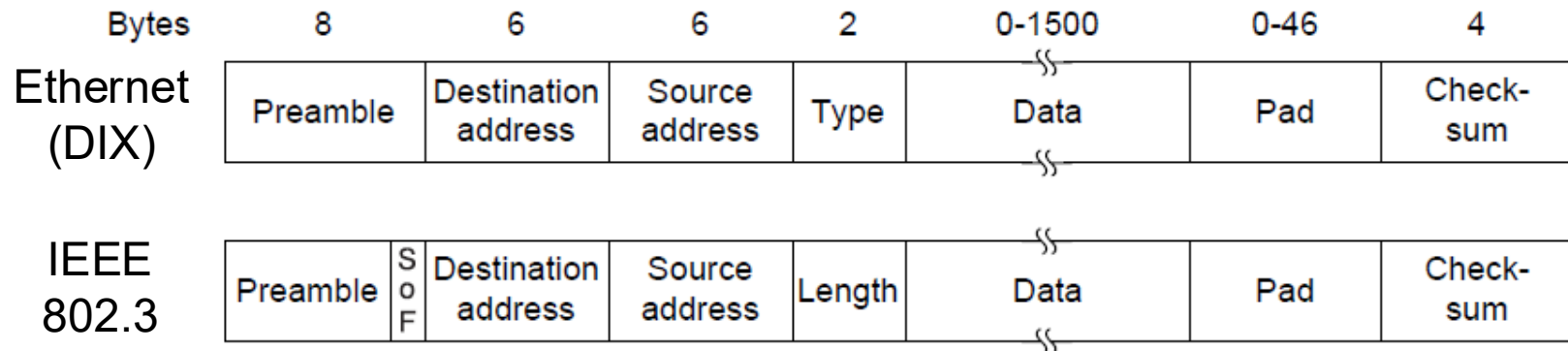
- Up to 10 Mbps, with Manchester encoding
- Hosts ran the classic Ethernet protocol for access



# Classic Ethernet (2) – MAC

MAC protocol is 1-persistent CSMA/CD (earlier)

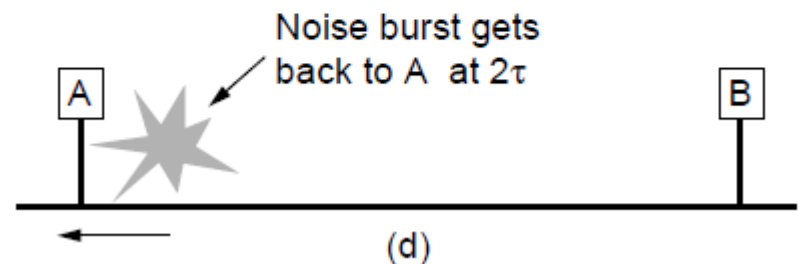
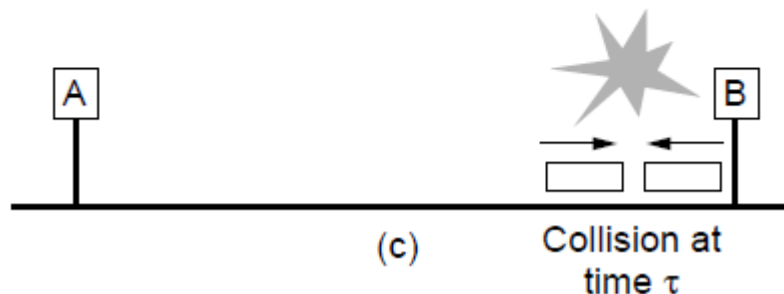
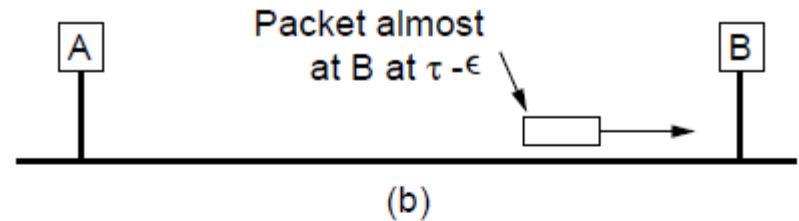
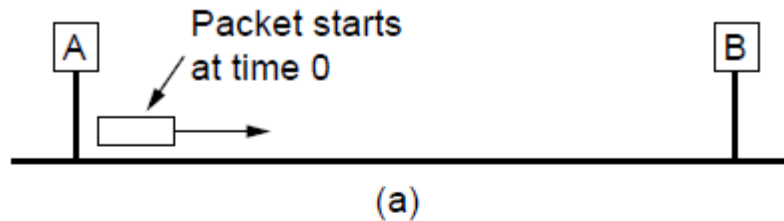
- Random delay (backoff) after collision is computed with BEB (Binary Exponential Backoff)
- Frame format is still used with modern Ethernet.



# Classic Ethernet (3) – MAC

Collisions can occur and take as long as  $2\tau$  to detect

- $\tau$  is the time it takes to propagate over the Ethernet
- Leads to minimum packet size for reliable detection

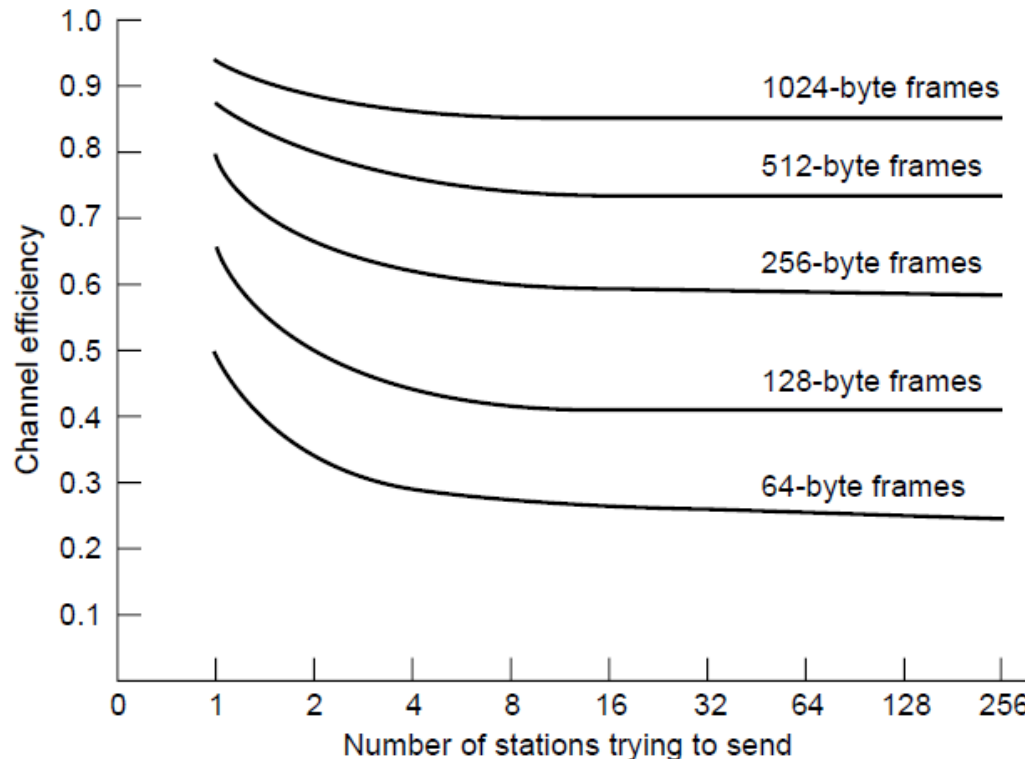




# Classic Ethernet (4) – Performance

Efficient for large frames, even with many senders

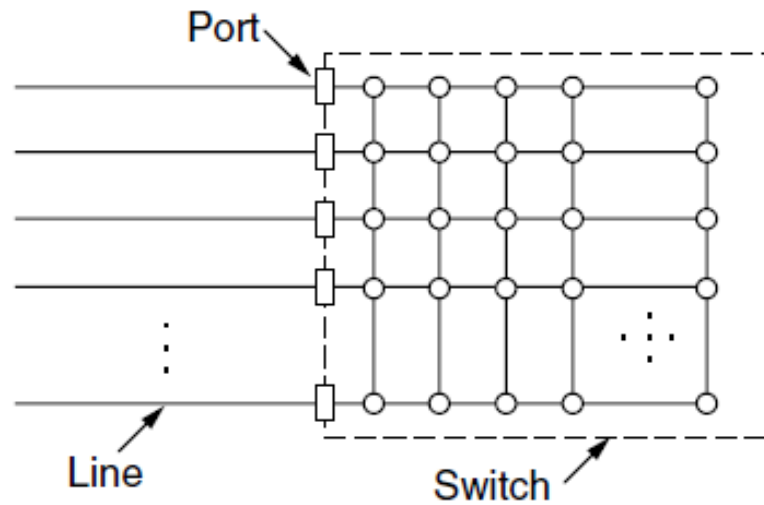
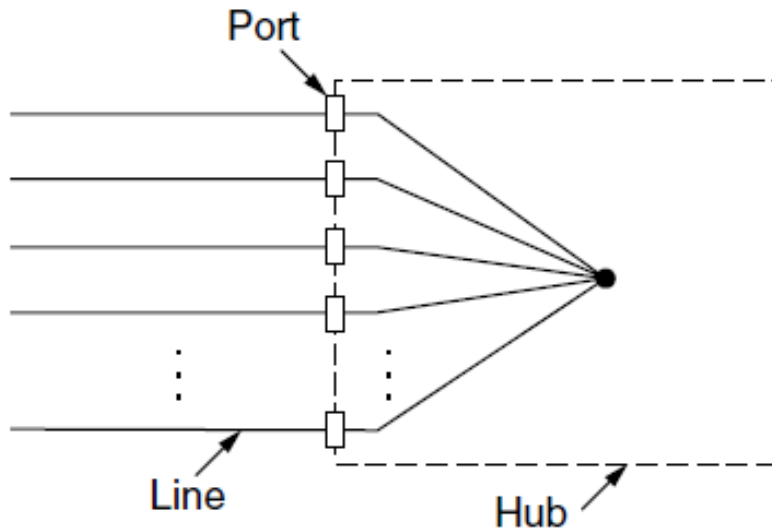
- Degrades for small frames (and long LANs)



10 Mbps Ethernet,  
64 byte min. frame

# Switched/Fast Ethernet (1)

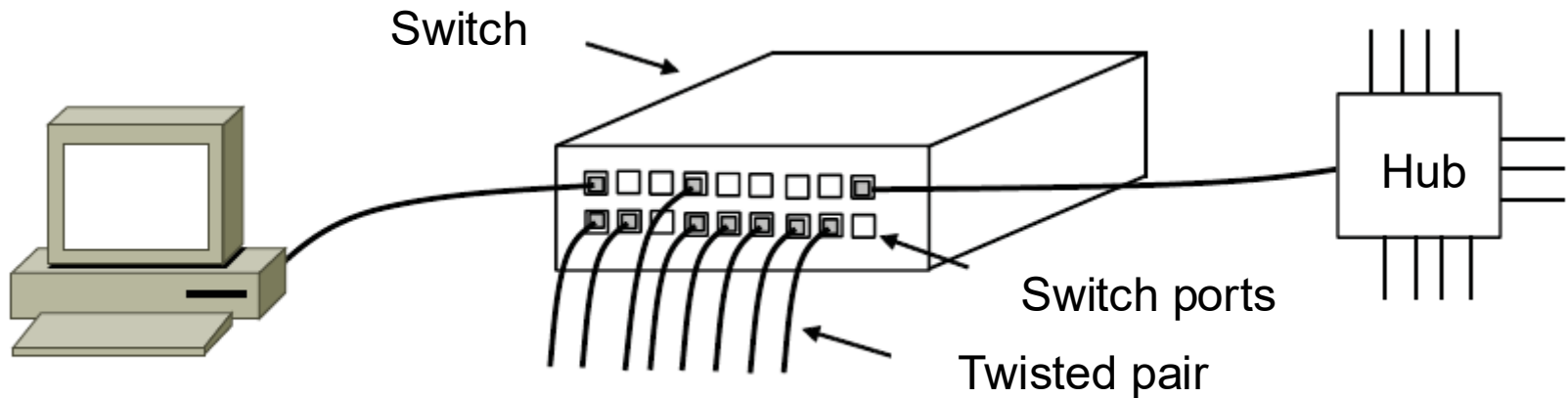
- Hubs wire all lines into a single CSMA/CD domain
- Switches isolate each port to a separate domain
  - Much greater throughput for multiple ports
  - No need for CSMA/CD with full-duplex lines



# Switched/Fast Ethernet (2)

Switches can be wired to computers, hubs and switches

- Hubs concentrate traffic from computers
- More on how to switch frames the in 4.8



# Switched/Fast Ethernet (3)

Fast Ethernet extended Ethernet from 10 to 100 Mbps

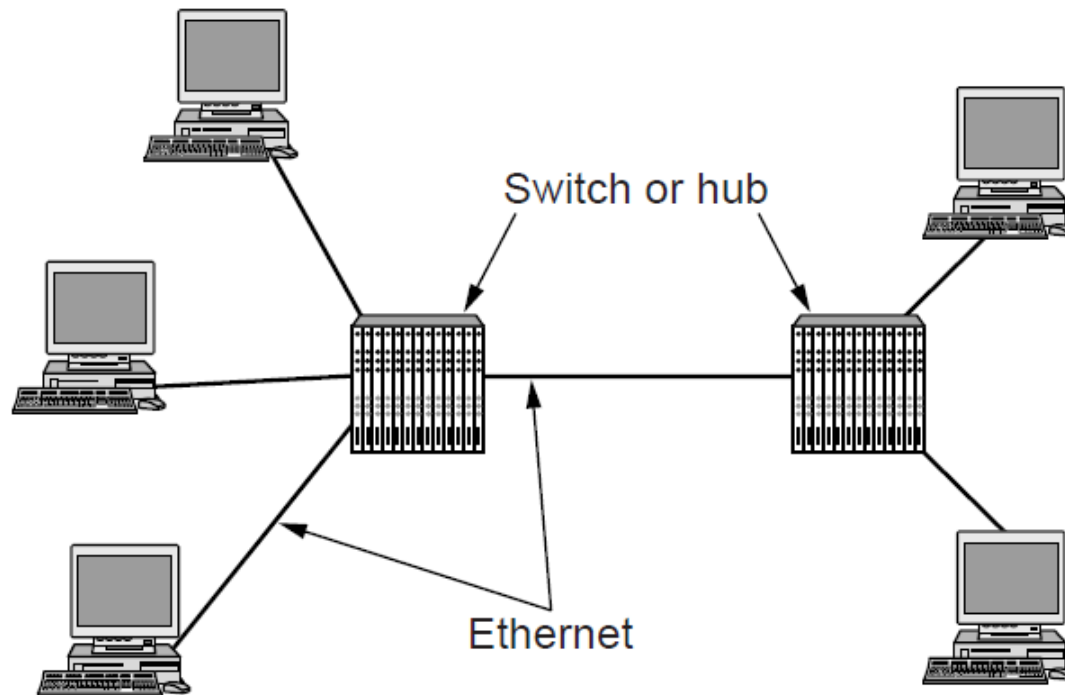
- Twisted pair (with Cat 5) dominated the market

| Name       | Cable        | Max. segment | Advantages                          |
|------------|--------------|--------------|-------------------------------------|
| 100Base-T4 | Twisted pair | 100 m        | Uses category 3 UTP                 |
| 100Base-TX | Twisted pair | 100 m        | Full duplex at 100 Mbps (Cat 5 UTP) |
| 100Base-FX | Fiber optics | 2000 m       | Full duplex at 100 Mbps; long runs  |

# Gigabit / 10 Gigabit Ethernet (1)

Switched Gigabit Ethernet is now the garden variety

- With full-duplex lines between computers/switches



# Gigabit / 10 Gigabit Ethernet (1)

- Gigabit Ethernet is commonly run over twisted pair

| Name        | Cable          | Max. segment | Advantages                                        |
|-------------|----------------|--------------|---------------------------------------------------|
| 1000Base-SX | Fiber optics   | 550 m        | Multimode fiber (50, 62.5 microns)                |
| 1000Base-LX | Fiber optics   | 5000 m       | Single (10 $\mu$ ) or multimode (50, 62.5 $\mu$ ) |
| 1000Base-CX | 2 Pairs of STP | 25 m         | Shielded twisted pair                             |
| 1000Base-T  | 4 Pairs of UTP | 100 m        | Standard category 5 UTP                           |

- 10 Gigabit Ethernet is being deployed where needed

| Name        | Cable             | Max. segment | Advantages                     |
|-------------|-------------------|--------------|--------------------------------|
| 10GBase-SR  | Fiber optics      | Up to 300 m  | Multimode fiber (0.85 $\mu$ )  |
| 10GBase-LR  | Fiber optics      | 10 km        | Single-mode fiber (1.3 $\mu$ ) |
| 10GBase-ER  | Fiber optics      | 40 km        | Single-mode fiber (1.5 $\mu$ ) |
| 10GBase-CX4 | 4 Pairs of twinax | 15 m         | Twinaxial copper               |
| 10GBase-T   | 4 Pairs of UTP    | 100 m        | Category 6a UTP                |

- 40/100 Gigabit Ethernet is under development

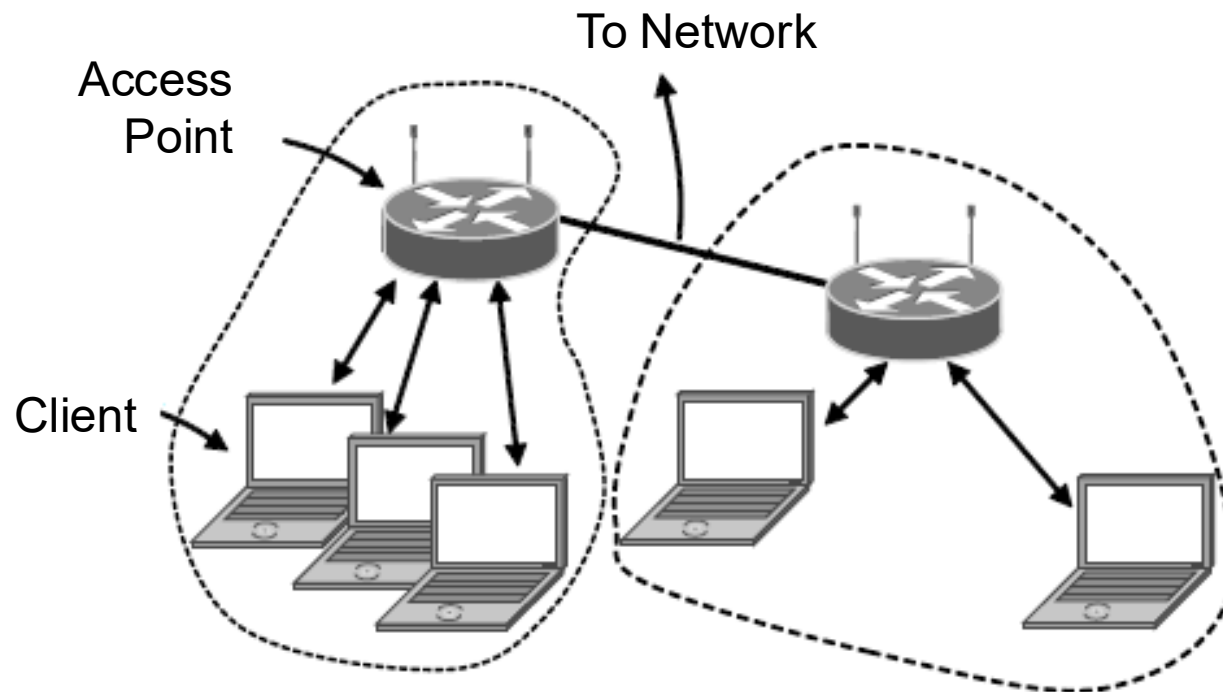
# Wireless LANs

- 802.11 architecture/protocol stack »
- 802.11 physical layer »
- 802.11 MAC »
- 802.11 frames »

# 802.11 Architecture/Protocol Stack (1)

Wireless clients associate to a wired AP (Access Point)

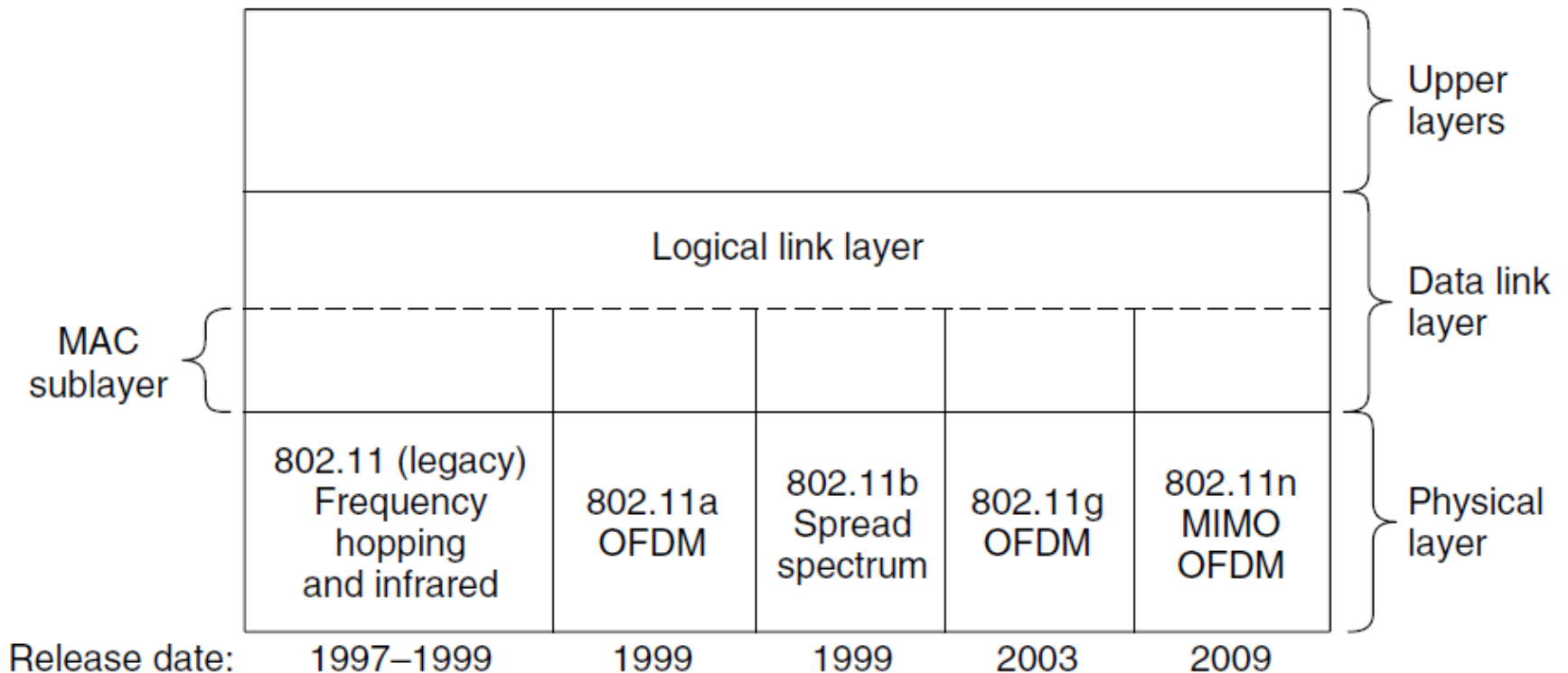
- Called infrastructure mode; there is also ad-hoc mode with no AP, but that is rare.





# 802.11 Architecture/Protocol Stack (2)

MAC is used across different physical layers



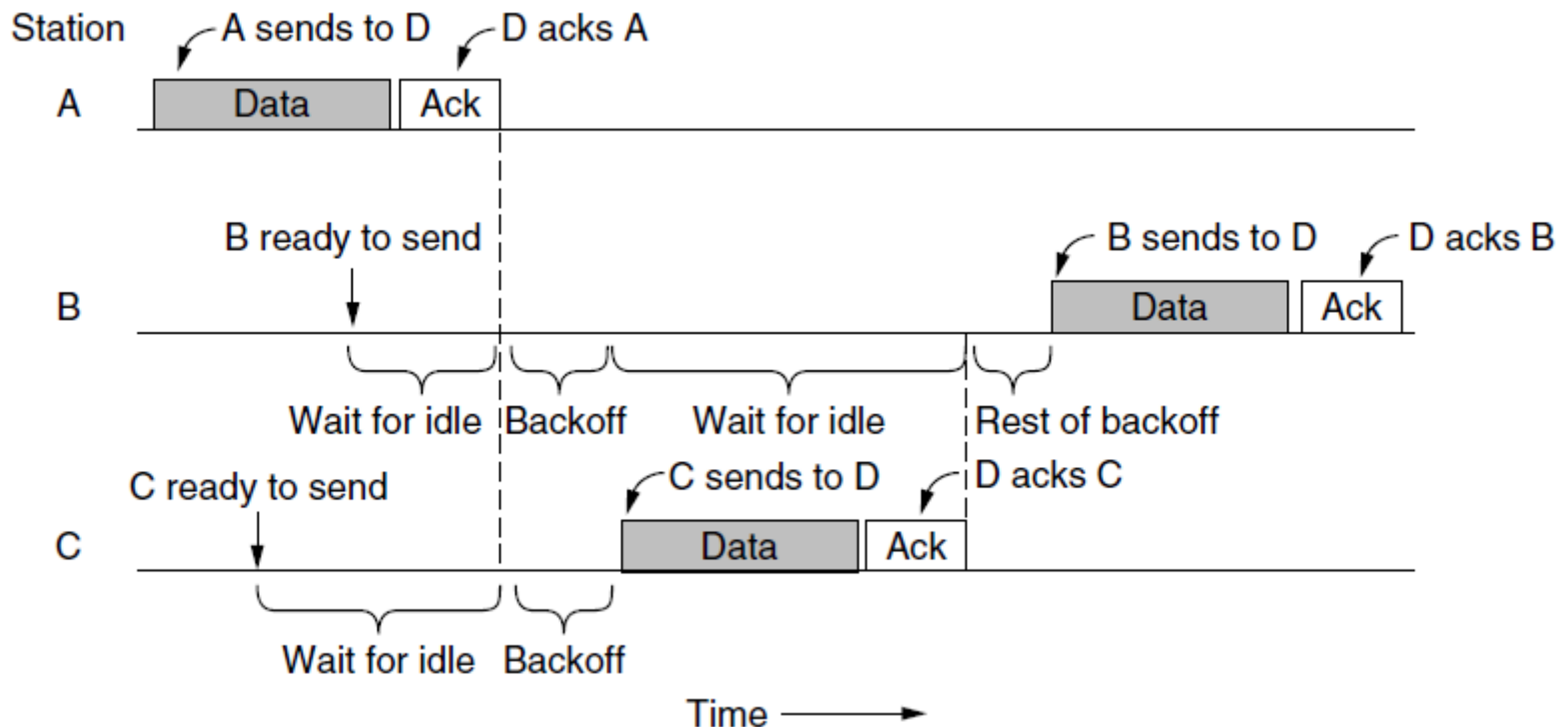
# 802.11 physical layer

- NICs are compatible with multiple physical layers
  - E.g., 802.11 a/b/g

| Name    | Technique                 | Max. Bit Rate |
|---------|---------------------------|---------------|
| 802.11b | Spread spectrum, 2.4 GHz  | 11 Mbps       |
| 802.11g | OFDM, 2.4 GHz             | 54 Mbps       |
| 802.11a | OFDM, 5 GHz               | 54 Mbps       |
| 802.11n | OFDM with MIMO, 2.4/5 GHz | 600 Mbps      |

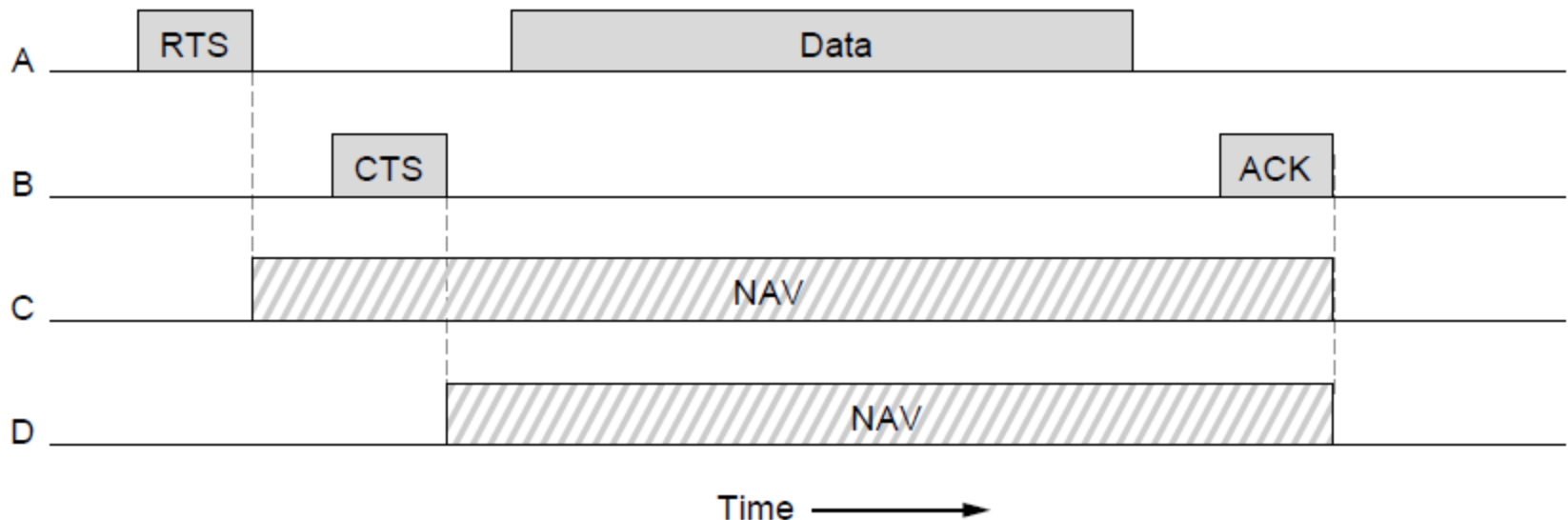
# 802.11 MAC (1)

- CSMA/CA inserts backoff slots to avoid collisions
- MAC uses ACKs/retransmissions for wireless errors



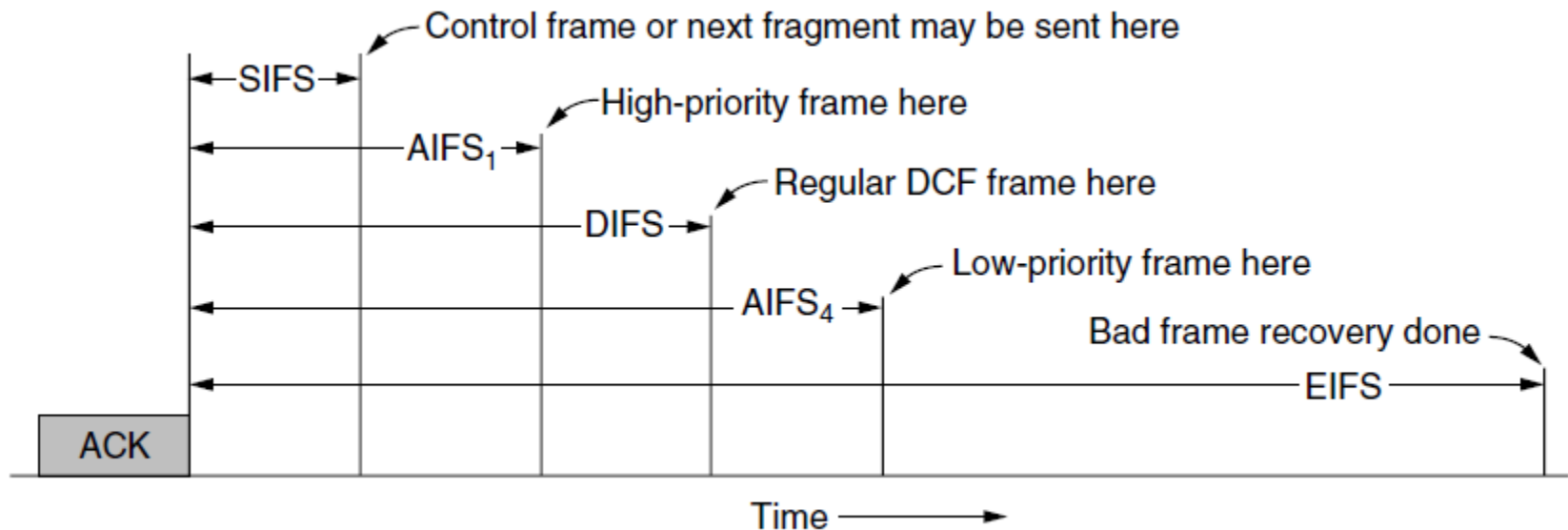
# 802.11 MAC (2)

Virtual channel sensing with the NAV and optional RTS/CTS (often not used) avoids hidden terminals



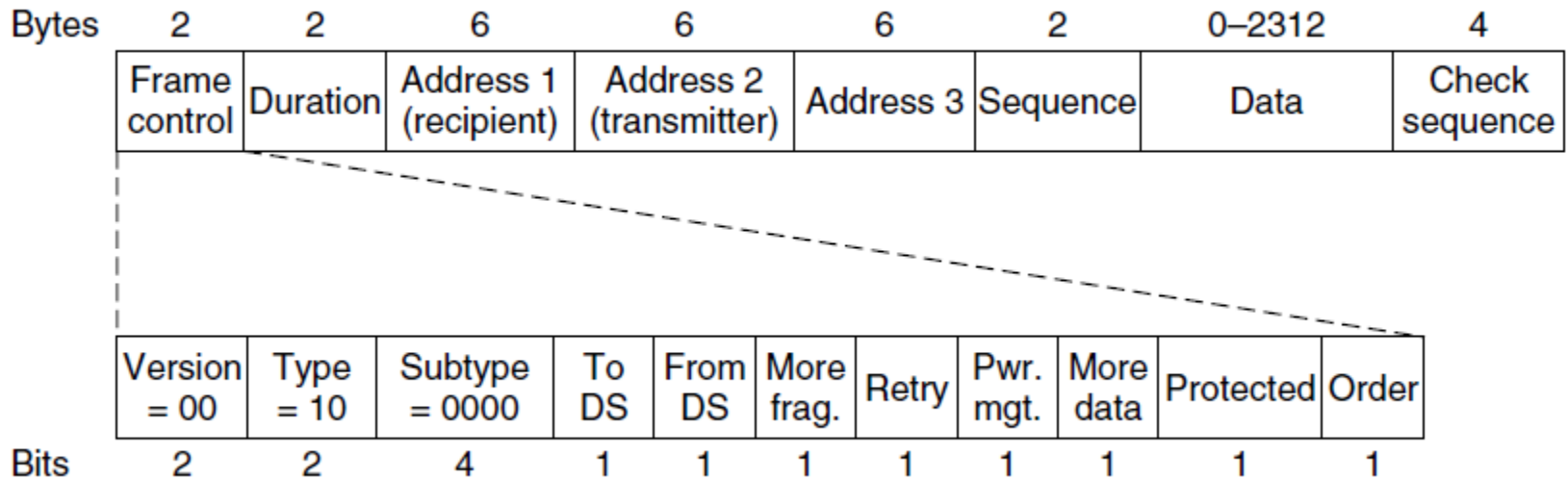
# 802.11 MAC (3)

- Different backoff slot times add quality of service
  - Short intervals give preferred access, e.g., control, VoIP
- MAC has other mechanisms too, e.g., power save



# 802.11 Frames

- Frames vary depending on their type (Frame control)
- Data frames have 3 addresses to pass via APs

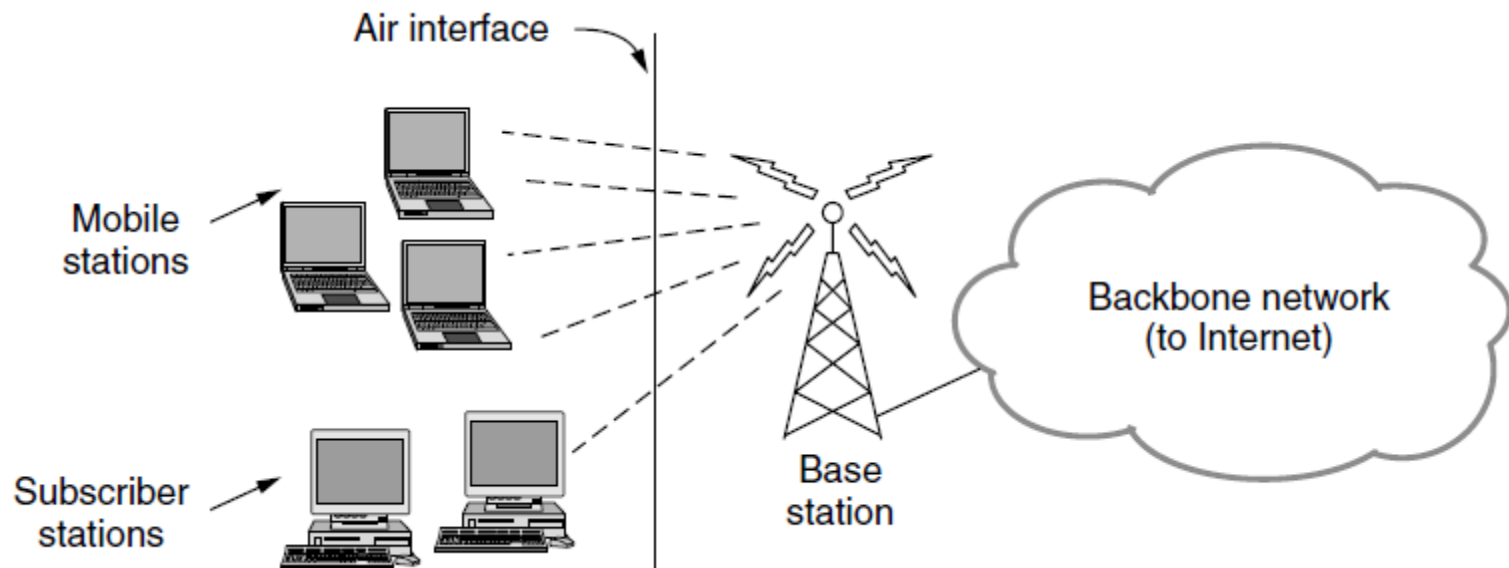


# Broadband Wireless

- 802.16 Architecture / Protocol Stack »
- 802.16 Physical Layer »
- 802.16 MAC »
- 802.16 Frames »

# 802.16 Architecture/Protocol Stack (1)

Wireless clients connect to a wired basestation (like 3G)

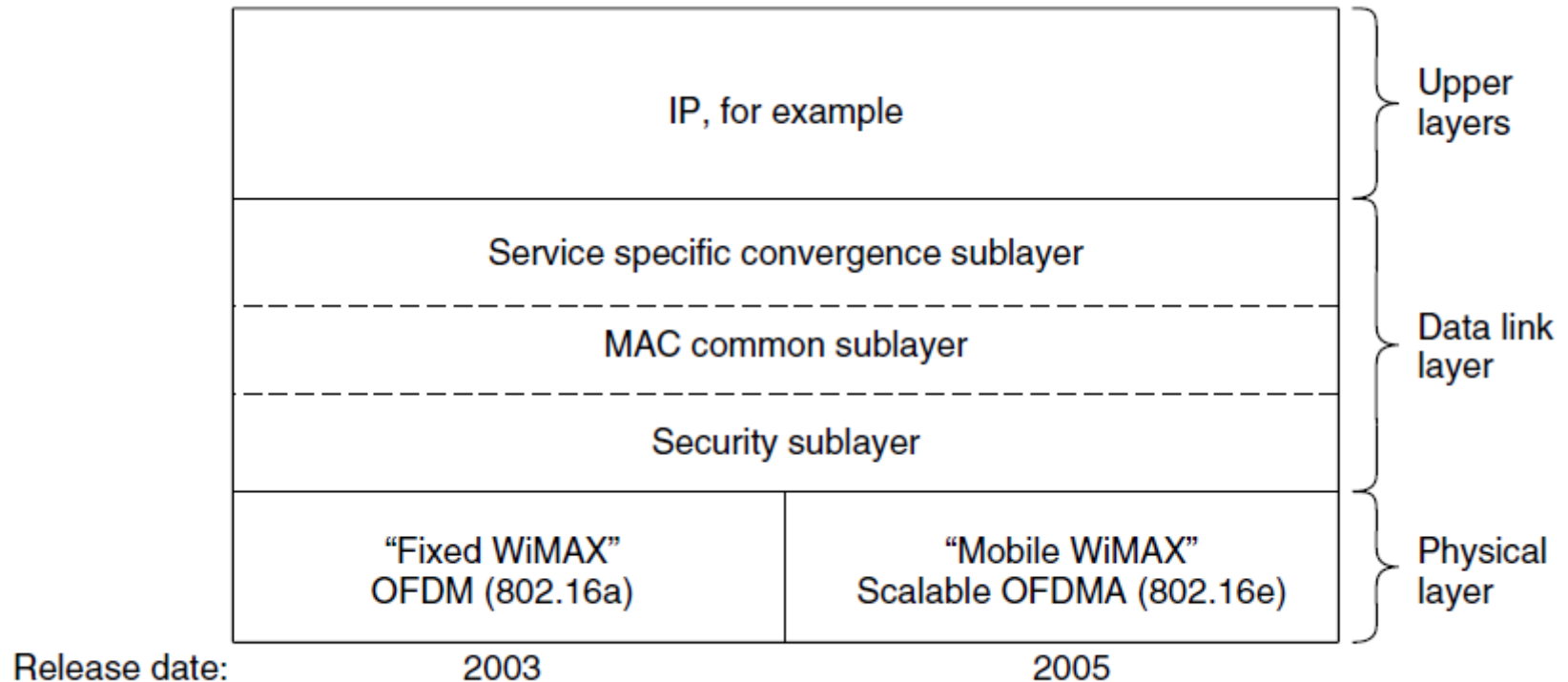




# 802.16 Architecture/Protocol Stack (2)

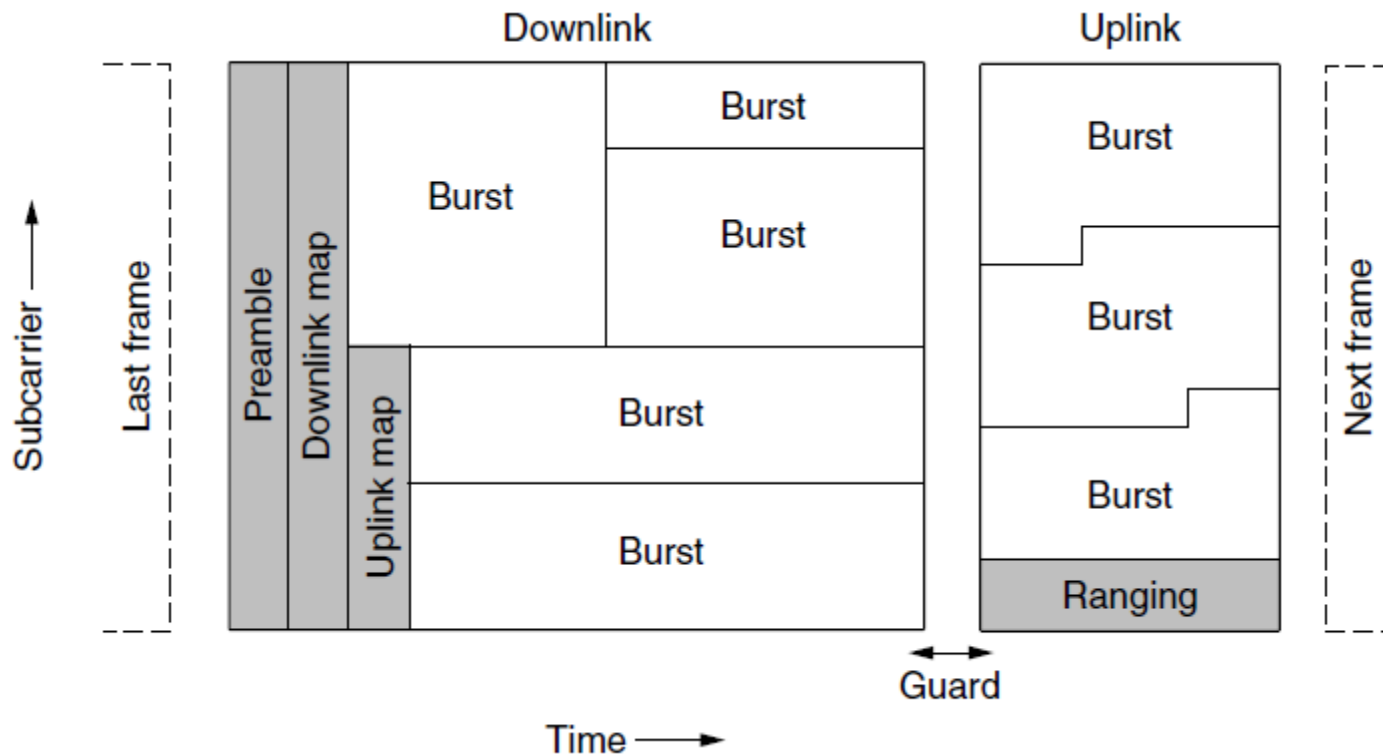
MAC is connection-oriented; IP is connectionless

- Convergence sublayer maps between the two



# 802.16 Physical Layer

Based on OFDM; base station gives mobiles bursts (subcarrier/time frame slots) for uplink and downlink



# 802.16 MAC

Connection-oriented with base station in control

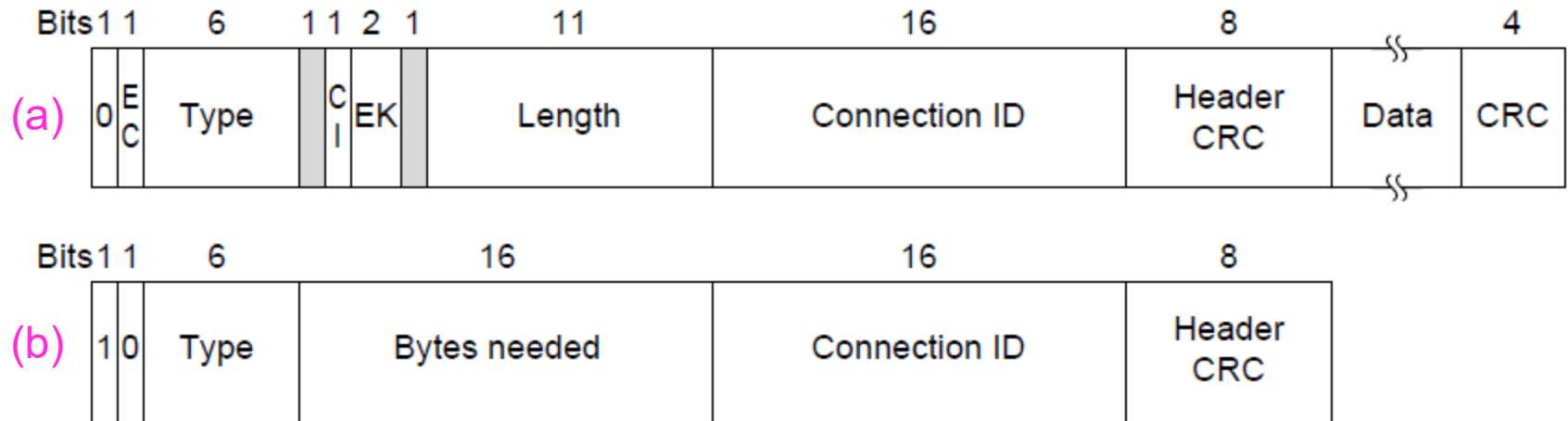
- Clients request the bandwidth they need

Different kinds of service can be requested:

- Constant bit rate, e.g., uncompressed voice
- Real-time variable bit rate, e.g., video, Web
- Non-real-time variable bit rate, e.g., file download
- Best-effort for everything else

# 802.16 Frames

- Frames vary depending on their type
- Connection ID instead of source/dest addresses



(a) A generic frame. (b) A bandwidth request frame

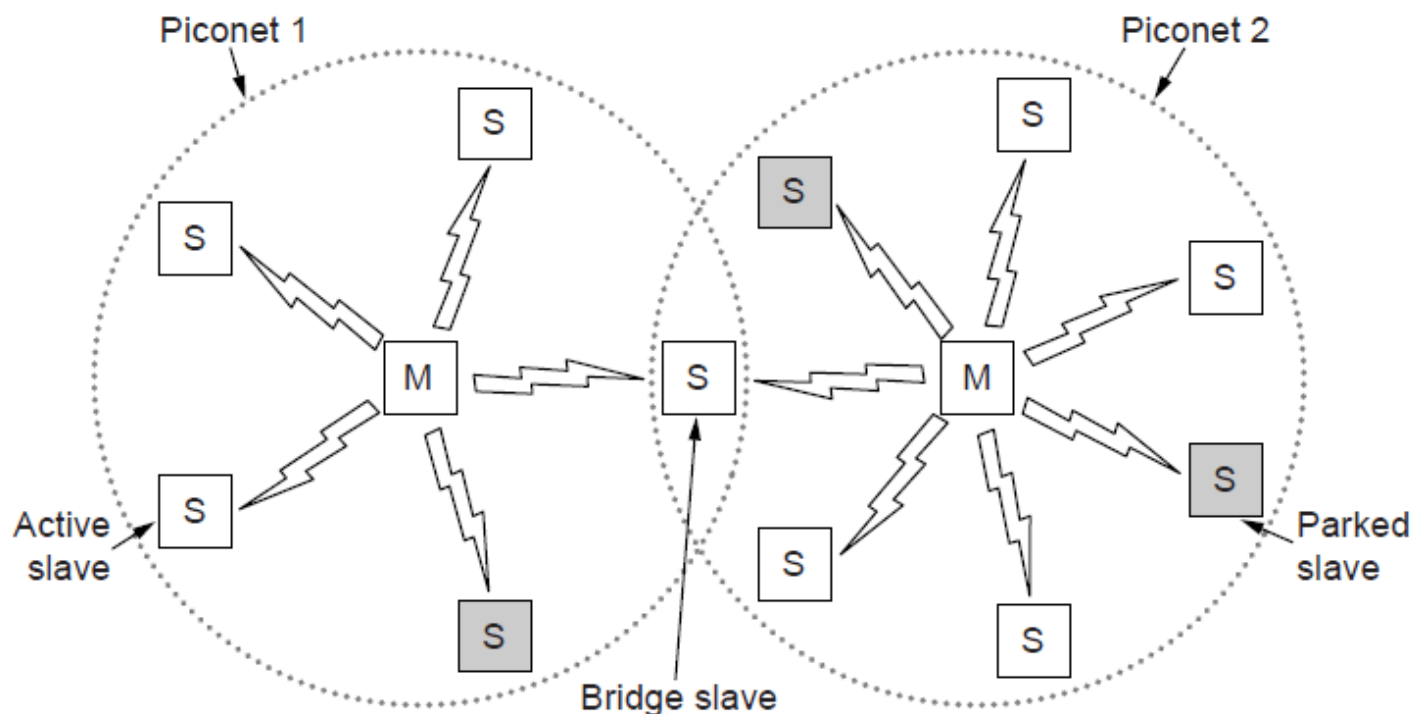
# Bluetooth

- Bluetooth Architecture »
- Bluetooth Applications / Protocol »
- Bluetooth Radio / Link Layers »
- Bluetooth Frames »

# Bluetooth Architecture

Piconet master is connected to slave wireless devices

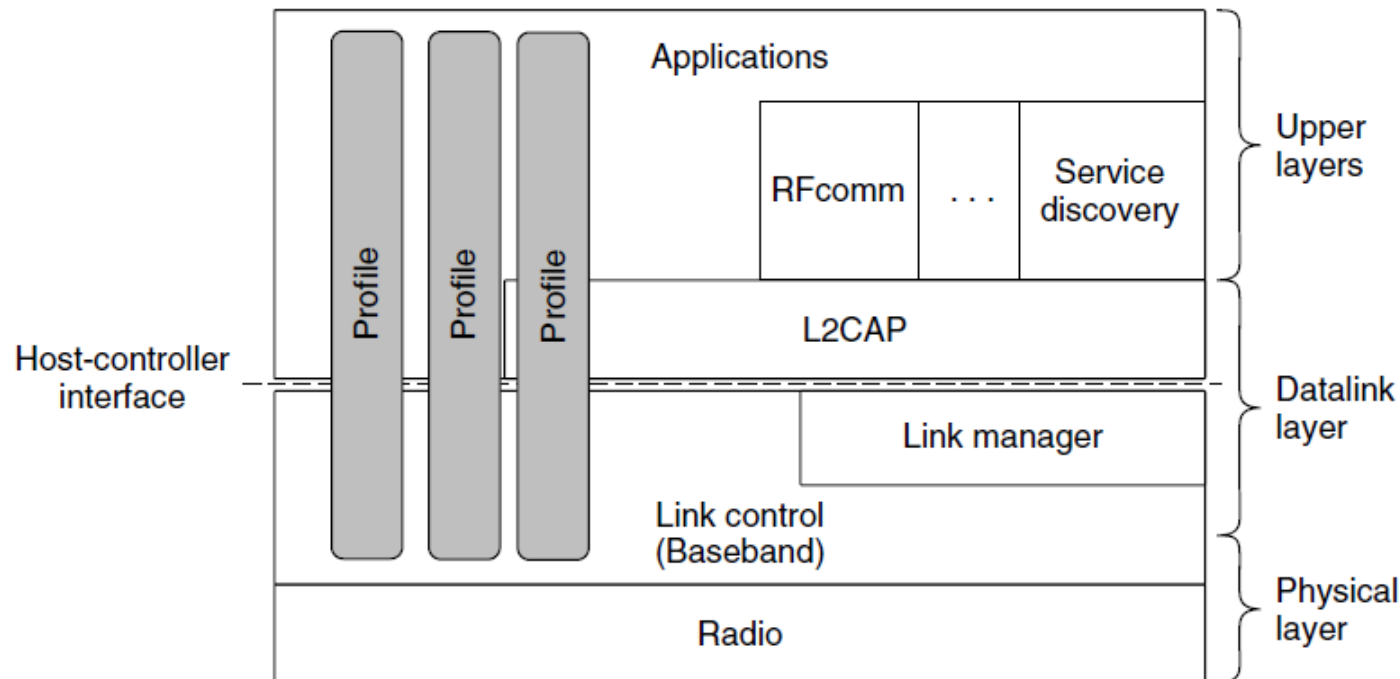
- Slaves may be asleep (parked) to save power
- Two piconets can be bridged into a scatternet



# Bluetooth Applications / Protocol Stack

Profiles give the set of protocols for a given application

- 25 profiles, including headset, intercom, streaming audio, remote control, personal area network, ...



# Bluetooth Radio / Link Layers

## Radio layer

- Uses adaptive frequency hopping in 2.4 GHz band

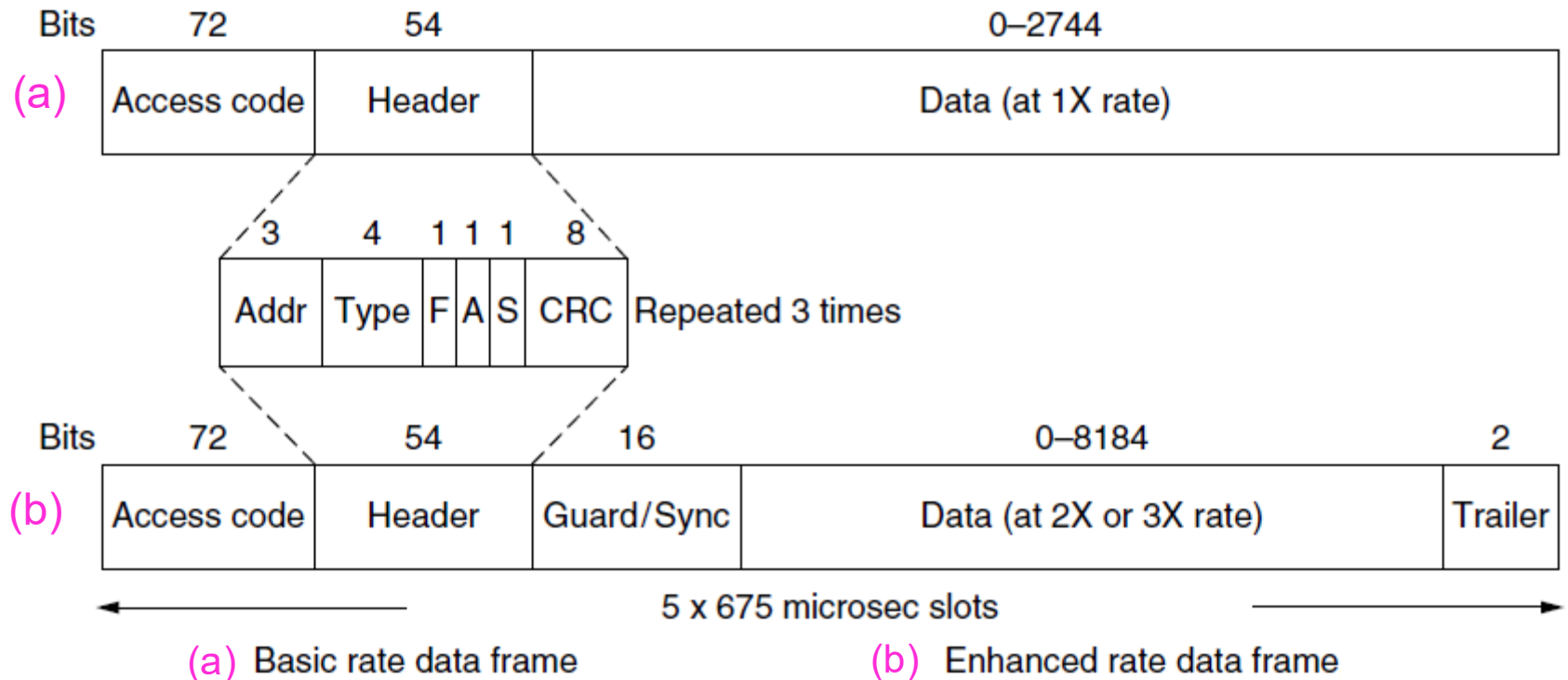
## Link layer

- TDM with timeslots for master and slaves
- Synchronous CO for periodic slots in each direction
- Asynchronous CL for packet-switched data
- Links undergo pairing (user confirms passkey/PIN) to authorize them before use



# Bluetooth Frames

Time is slotted; enhanced data rates send faster but for the same time; addresses are only 3 bits for 8 devices

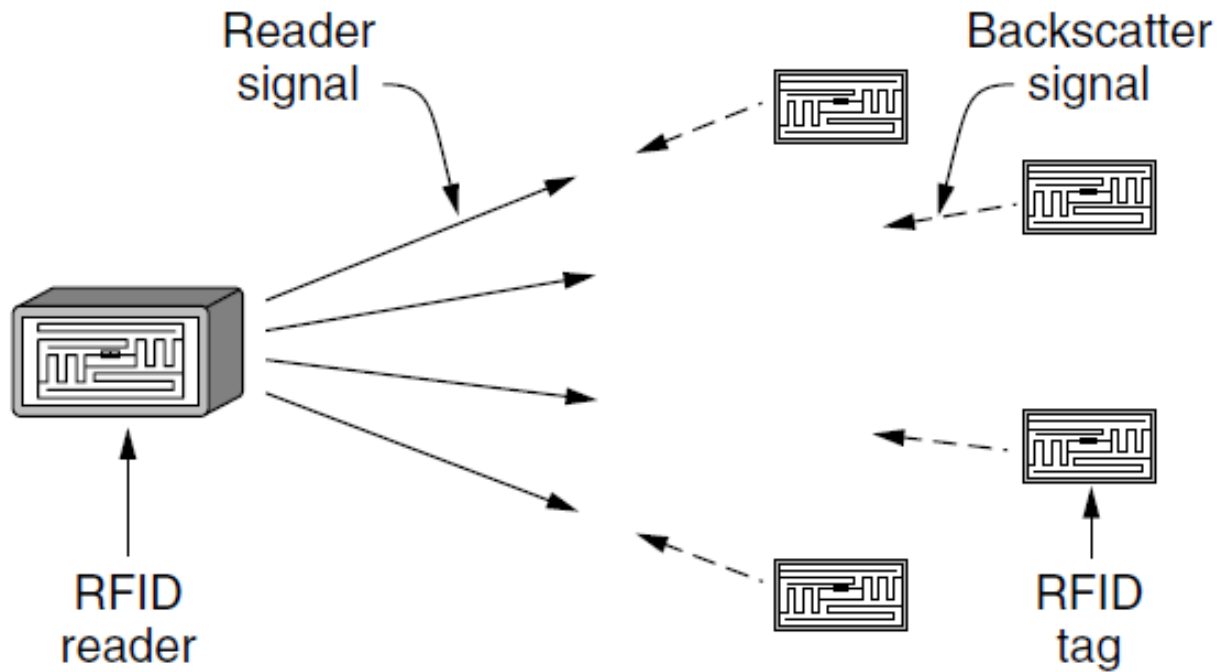


# RFID

- Gen 2 Architecture »
- Gen 2 Physical Layer »
- Gen 2 Tag Identification Layer »
- Gen 2 Frames »

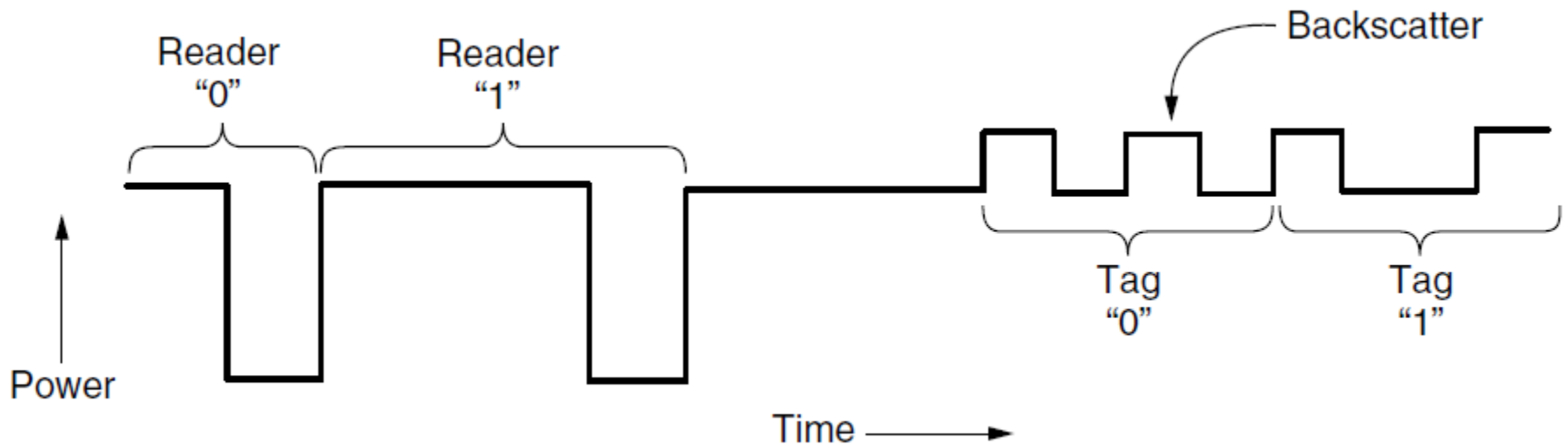
# Gen 2 Architecture

Reader signal powers tags; tags reply with backscatter



# Gen 2 Physical Layer

- Reader uses duration of on period to send 0/1
- Tag backscatters reader signal in pulses to send 0/1



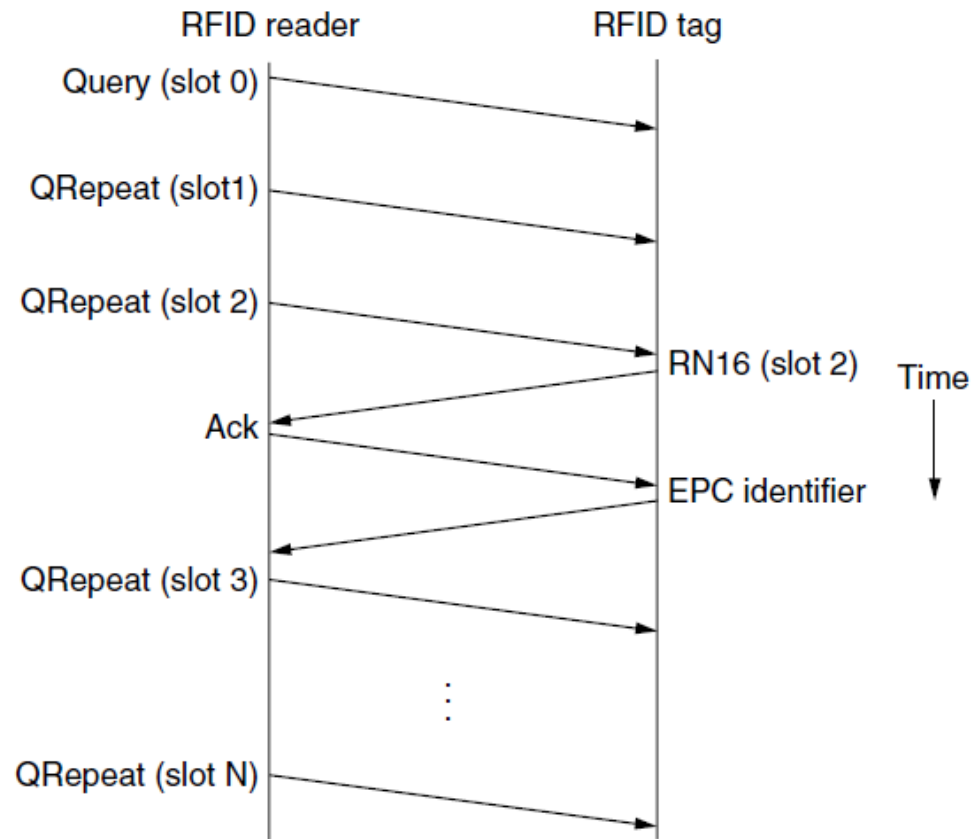
# Gen 2 Tag Identification Layer

Reader sends query and sets slot structure

Tags reply (RN16) in a random slot; may collide

Reader asks one tag for its identifier (ACK)

Process continues until no tags are left



# Gen 2 Frames

- Reader frames vary depending on type (Command)
  - Query shown below, has parameters and error detection
- Tag responses are simply data
  - Reader sets timing and knows the expected format



Query message

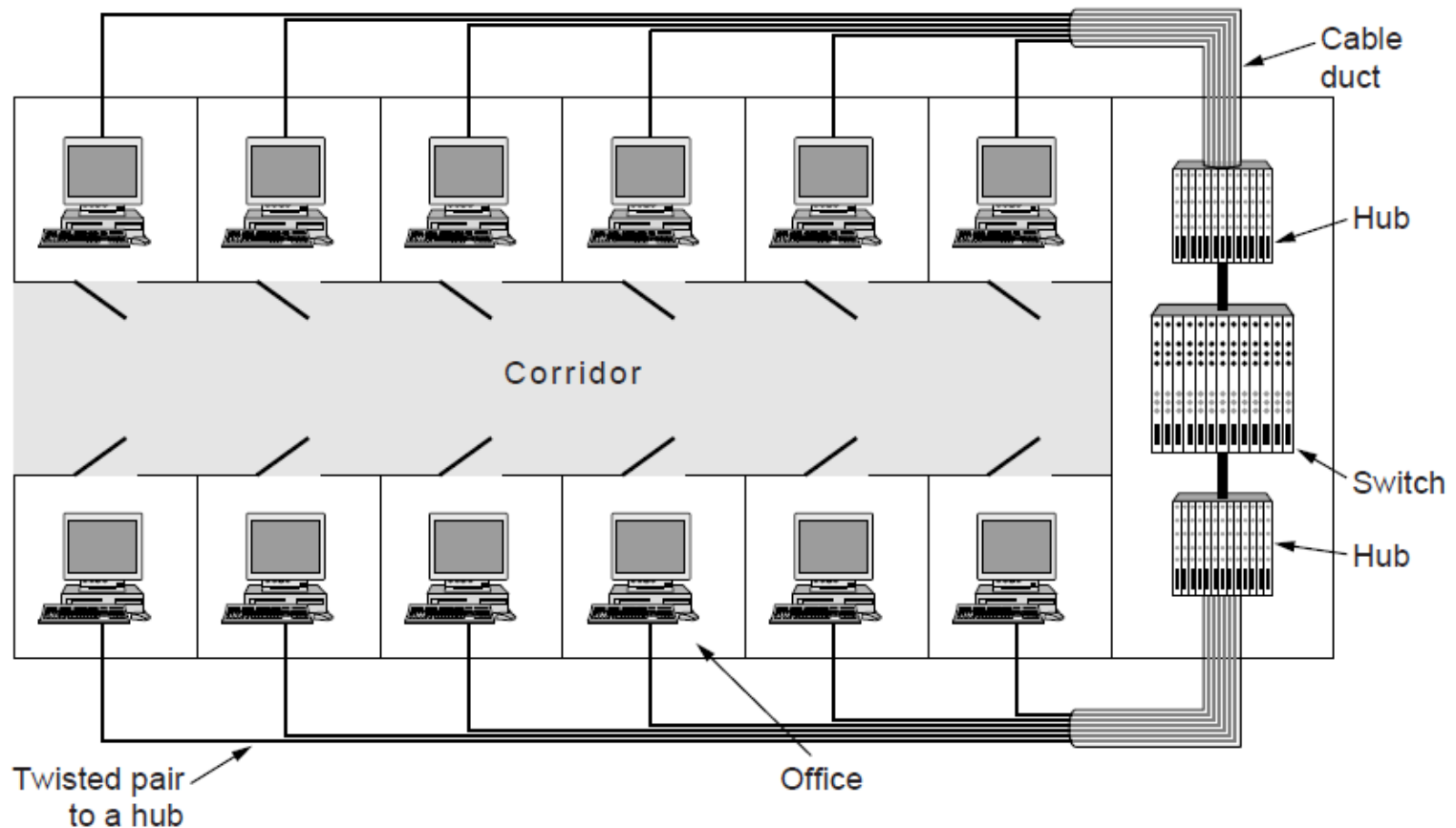
# Data Link Layer Switching

- Uses of Bridges »
- Learning Bridges »
- Spanning Tree »
- Repeaters, hubs, bridges, .., routers, gateways »
- Virtual LANs »

# Uses of Bridges

Common setup is a building with centralized wiring

- Bridges (switches) are placed in or near wiring closets

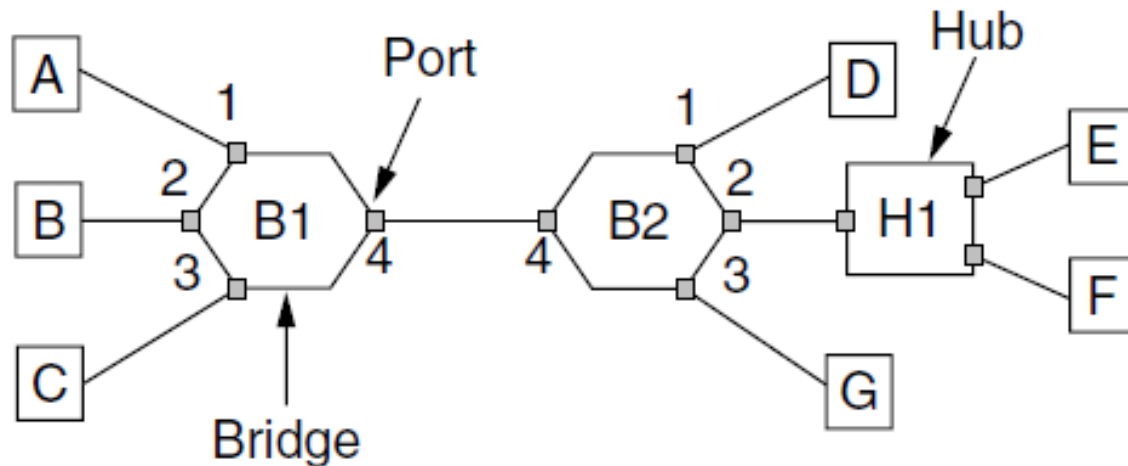




# Learning Bridges (1)

A bridge operates as a switched LAN (not a hub)

- Computers, bridges, and hubs connect to its ports



# Learning Bridges (2)

Backward learning algorithm picks the output port:

- Associates source address on frame with input port
- Frame with destination address sent to learned port
- Unlearned destinations are sent to all other ports

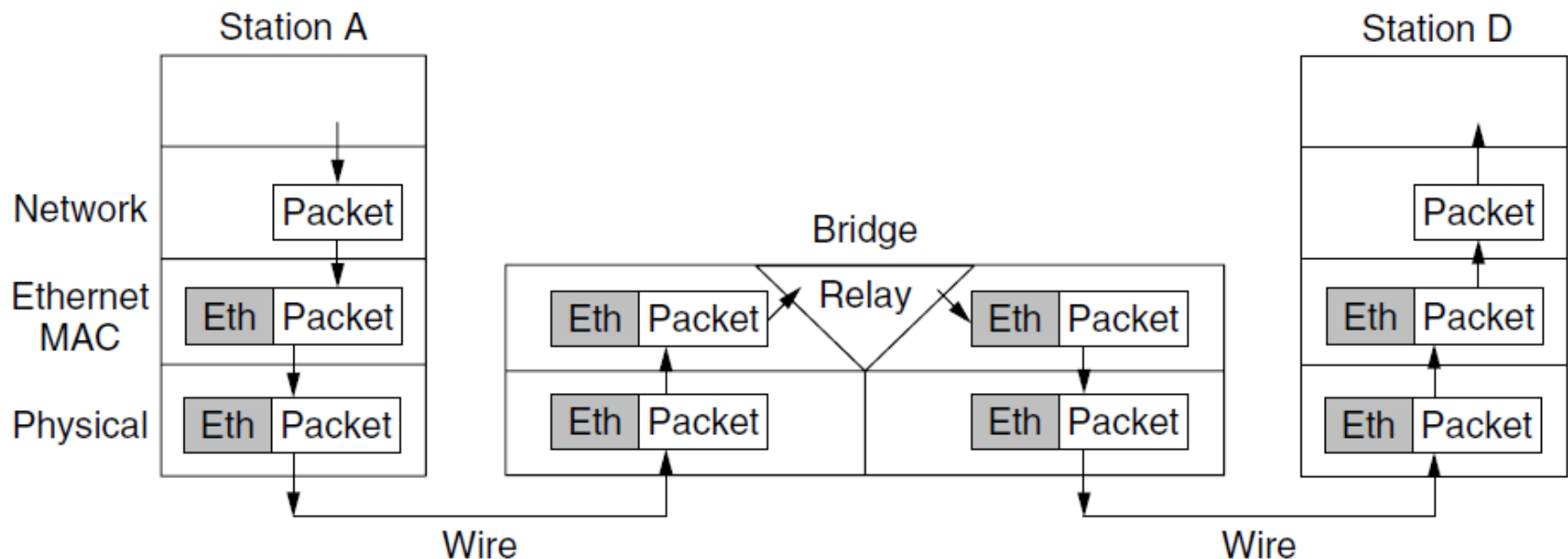
Needs no configuration

- Forget unused addresses to allow changes
- Bandwidth efficient for two-way traffic

# Learning Bridges (3)

Bridges extend the Link layer:

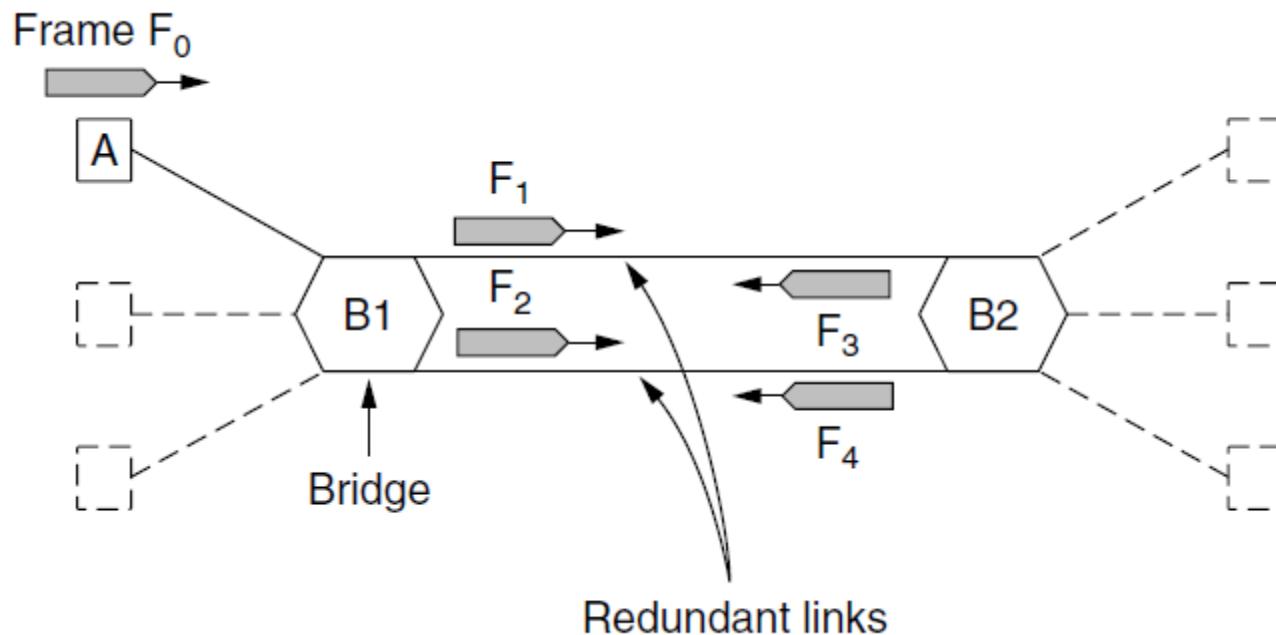
- Use but don't remove Ethernet header/addresses
- Do not inspect Network header



# Spanning Tree (1) – Problem

Bridge topologies with loops and only backward learning will cause frames to circulate for ever

- Need spanning tree support to solve problem



# Spanning Tree (2) – Algorithm

- Subset of forwarding ports for data is use to avoid loops
- Selected with the spanning tree distributed algorithm by Perlman

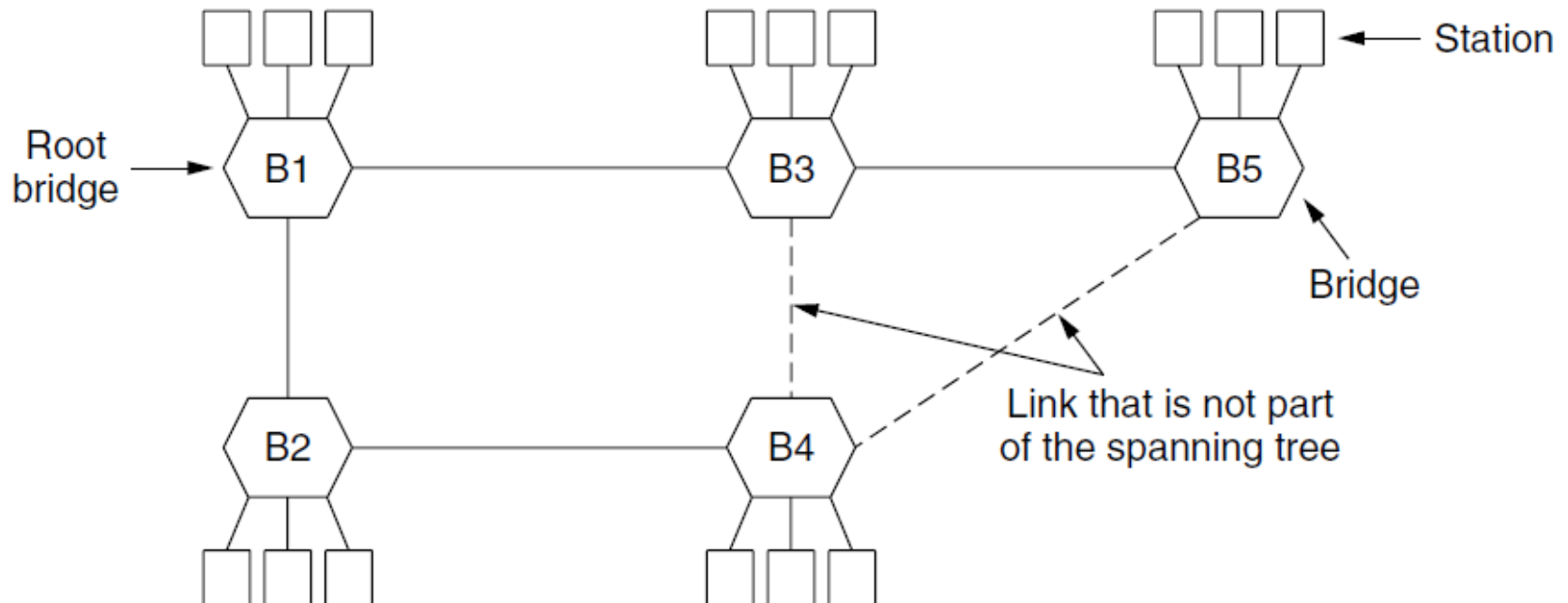
*I think that I shall never see  
A graph more lovely than a tree.  
A tree whose crucial property  
Is loop-free connectivity.  
A tree which must be sure to span.  
So packets can reach every LAN.  
First the Root must be selected  
By ID it is elected.  
Least cost paths from Root are traced  
In the tree these paths are placed.  
A mesh is made by folks like me  
Then bridges find a spanning tree.*

– Radia Perlman, 1985.

# Spanning Tree (3) – Example

After the algorithm runs:

- B1 is the root, two dashed links are turned off
- B4 uses link to B2 (lower than B3 also at distance 1)
- B5 uses B3 (distance 1 versus B4 at distance 2)



# Repeaters, Hubs, Bridges, Switches, Routers, & Gateways

Devices are named according to the layer they process

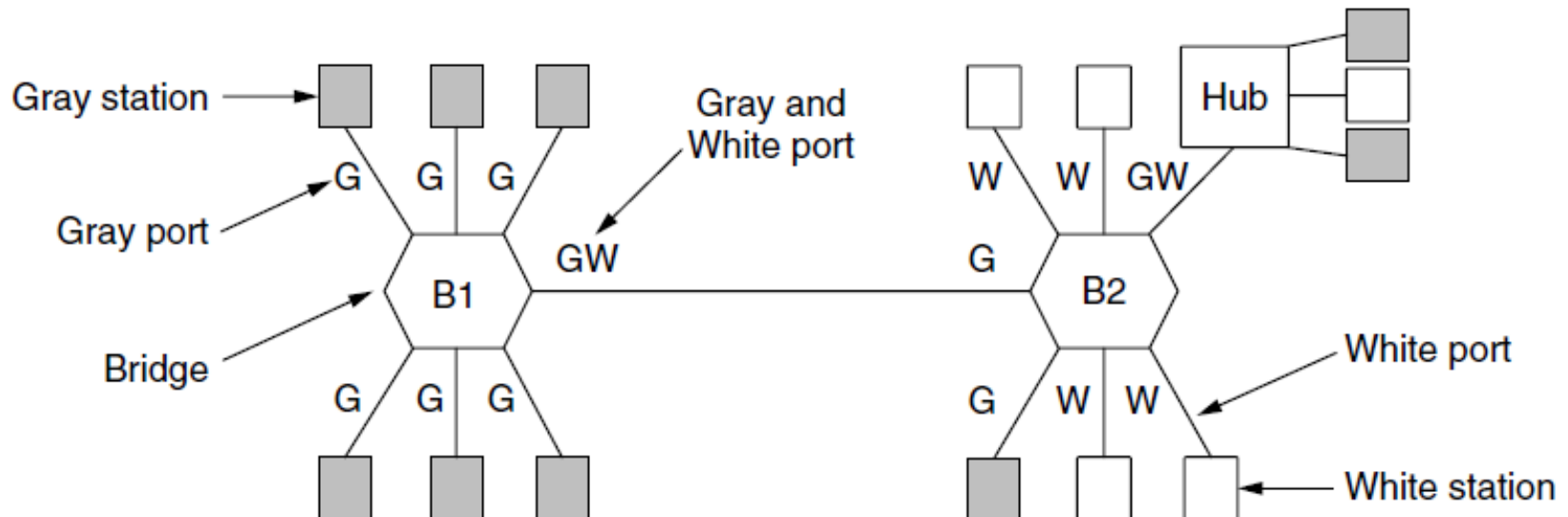
- A bridge or LAN switch operates in the Link layer

|                   |                     |
|-------------------|---------------------|
| Application layer | Application gateway |
| Transport layer   | Transport gateway   |
| Network layer     | Router              |
| Data link layer   | Bridge, switch      |
| Physical layer    | Repeater, hub       |

# Virtual LANs (1)

VLANs (Virtual LANs) splits one physical LAN into multiple logical LANs to ease management tasks

- Ports are “colored” according to their VLAN

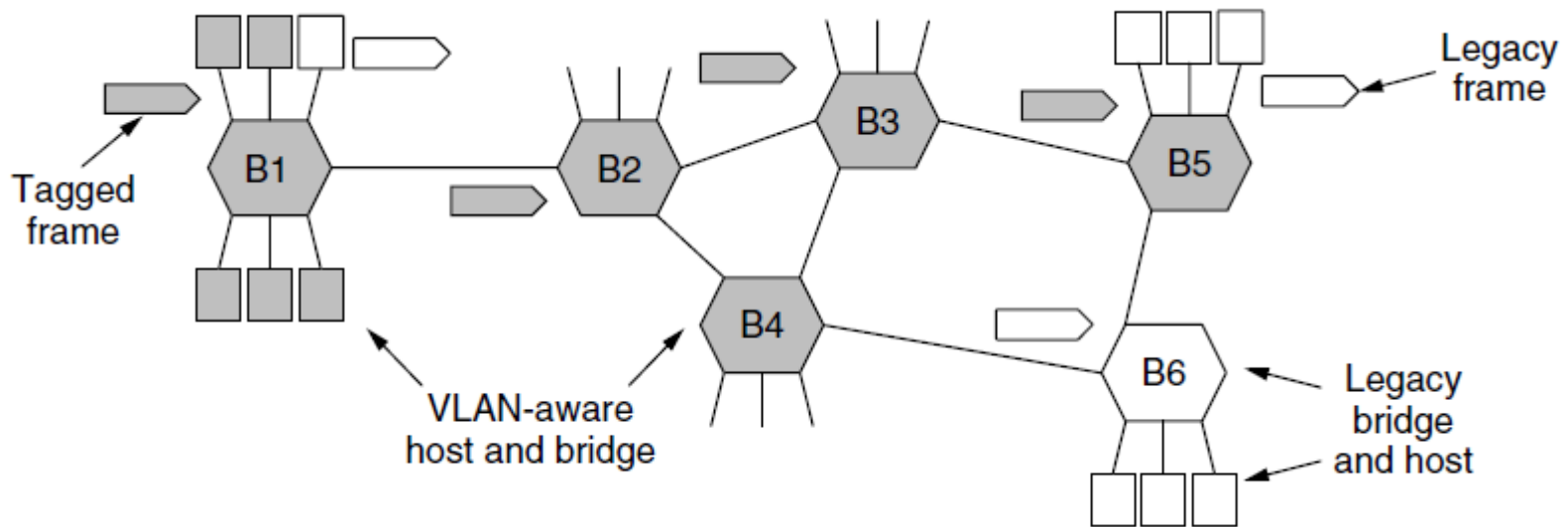




# Virtual LANs (2) – IEEE 802.1Q

Bridges need to be aware of VLANs to support them

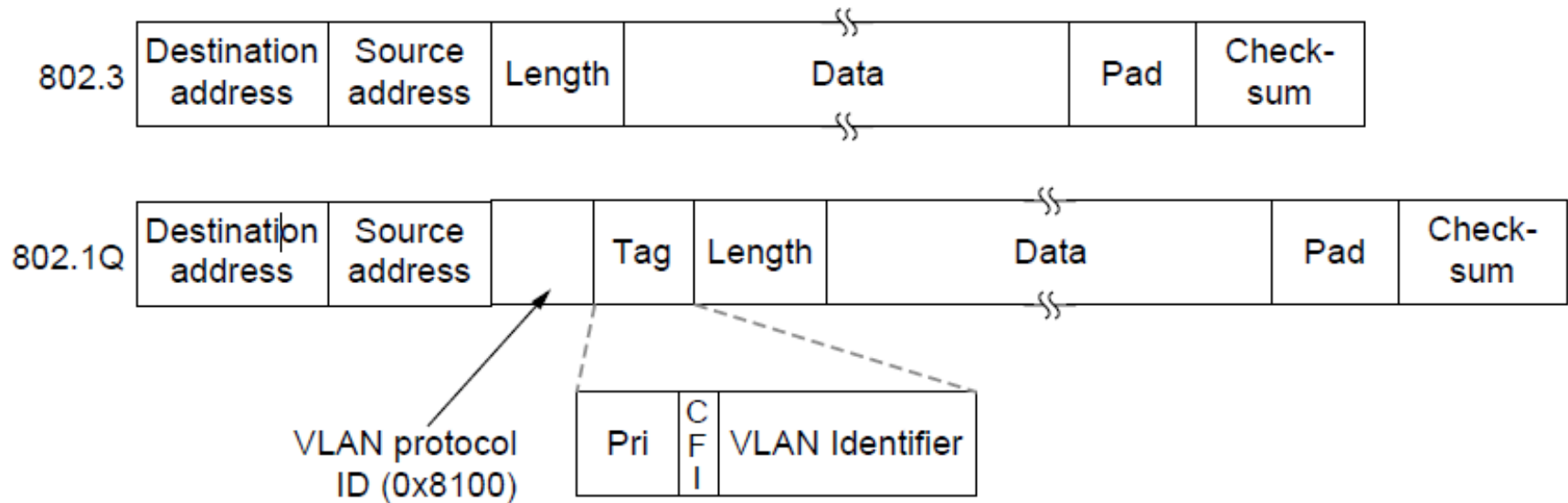
- In 802.1Q, frames are tagged with their “color”
- Legacy switches with no tags are supported



# Virtual LANs (3) – IEEE 802.1Q

802.1Q frames carry a color tag (VLAN identifier)

- Length/Type value is 0x8100 for VLAN protocol



End

Chapter 4